Thadomal Shahani Engineering College

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CERTIFICATE

Certify that Mr./Miss Asmit Pandey

of AIDS Department, Semester V with

Roll No 2401072 has completed a course of the necessary

experiments in the subject INNOC under my

supervision in the **Thadomal Shahani Engineering College**

Laboratory in the year 2025 - 2026

Teacher In- Charge

Head of the Department

Date 29/oct/25

Principal

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THAKUR SHAHANI
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Experiment No. 01

Title:

Study of Electromagnetic Spectrum Allocation and Wireless Bands Used in Communication

1. Aim

To study the electromagnetic spectrum, identify the radio frequency (RF) bands used for wireless communication, and understand their allocation to various services.

Specific Objectives:

- To understand the classification of the electromagnetic spectrum from ELF to Gamma rays.
- To identify the RF portion of the spectrum used for wireless communication.
- To study the major frequency bands (VLF to EHF) and their typical applications.
- To examine the concept of licensed vs. unlicensed spectrum and the role of regulatory bodies.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 1: Basics of Wireless Communication
Theory Section(s)	Wireless Transmission Principles / Frequency Spectrum Allocation
Course Outcome	CO1
Laboratory Outcome	LO1

3. Theoretical Background

3.1 Introduction

The foundation of all wireless communication is the electromagnetic spectrum. Every wireless system — from mobile phones and Wi-Fi to satellite communication and radar — relies on a specific range of frequencies to transmit information. The electromagnetic spectrum encompasses all types of electromagnetic radiation, ranging from extremely low frequencies (ELF) used for submarine communication to high-energy gamma rays used in medical treatments.

For wireless communication, the most relevant portion is the Radio Frequency (RF) spectrum, which spans from approximately 3 kHz to 300 GHz. Since this spectrum is a finite natural resource, its allocation and management are essential to prevent interference and ensure that different services can coexist without disruption.

3.2 Fundamental Concepts

a) ***Electromagnetic Spectrum:*** The complete range of electromagnetic waves arranged by frequency or wavelength. All waves travel at the speed of light ($\approx 3 \times 10^8$ m/s) but differ in frequency, wavelength, and energy.

b) ***Radio Frequency (RF) Spectrum:*** The portion of the electromagnetic spectrum from 3 kHz to 300 GHz, used for wireless communication.

c) ***Frequency Allocation:*** The process of assigning specific frequency bands to different wireless services (e.g., cellular, broadcasting, satellite).

d) ***Licensed Spectrum:*** Frequency bands that are assigned exclusively to specific operators through government auctions. Cellular networks (2G, 3G, 4G, 5G) operate in licensed spectrum to guarantee Quality of Service (QoS) and controlled interference.

e) ***Unlicensed Spectrum:*** Frequency bands that can be used without a license, provided devices comply with power limits. The ISM bands (2.4 GHz, 5 GHz) are the most prominent examples, used by Wi-Fi, Bluetooth, and Zigbee.

3.3 Working Principle

Each wireless technology is assigned a suitable frequency band depending on propagation characteristics, bandwidth requirement, coverage and application. Lower frequencies provide longer coverage while higher frequencies generally provide larger bandwidth.

Spectrum allocation operates on strict regulatory frameworks to ensure fair access and interference-free operation. Globally, this is managed by the International Telecommunication Union (ITU), while nationally, bodies like the Telecom Regulatory Authority of India (TRAI) oversee distribution. Operators are granted exclusive rights to specific frequency channels through a licensing paradigm, while certain bands are left open for general use under strict power limits.

3.4 Parameters to be Observed/Measured

- ***Frequency Band:*** The specific range of frequencies (e.g., VHF, UHF, SHF).
- ***Frequency Range:*** The start and end frequencies of a band.
- ***Typical Applications:*** The services that operate in each band.
- ***Regulatory Status:*** Whether the band is licensed or unlicensed.

3.5 Practical Applications

AM/FM broadcasting, television, cellular communication (2G–5G), Wi-Fi, Bluetooth, satellite communication, GPS, radar and IoT devices.

4. Requirements

Hardware: PC / Laptop with internet access

Software / Online Resources: Web browser to access ITU and TRAI websites.

5. Experimental Setup

Since this is an analytical study experiment, the setup involves accessing hierarchical regulatory portals rather than physical circuitry. Reference charts/online spectrum allocation table → Student observation → Recording sheet.

6. Procedure

1. Study the theoretical background provided in this manual (Section 3).
2. Search online for "electromagnetic spectrum chart" or "radio frequency bands" using any search engine.
3. Observe how different sources present the spectrum information.
4. Fill in the observation tables based on your understanding from the manual and online research.
5. Answer the analysis questions.

7. Observations

Table 1: Frequency Bands and Applications

Band	Frequency Range	Typical Applications
VLF		
LF		
MF		
HF		
VHF		
UHF		
SHF		
EHF		

Table 2: Indian Spectrum Allocation

Band	Typical Technology	Operators (Examples)
900 MHz		
1800 MHz		
2100 MHz		
2300 MHz		
3300–3600 MHz		

Table 3: Licensed vs Unlicensed Spectrum

Parameter	Licensed Spectrum	Unlicensed Spectrum
Permission Required		
Cost		
Interference		
Quality of Service		
Examples		

8. Analysis and Discussion

1. From the frequency bands table, (Table 1), observe the pattern between frequency and range. What general relationship do you observe?

[Write your answer here]

2. Based on Table 2 (Indian Spectrum Allocation), which frequency band is used by the maximum number of operators?

[Write your answer here]

3. From Table 3 (Licensed vs Unlicensed Spectrum), what is the key difference in terms of cost and quality of service? Which type would you recommend for critical services like emergency communication?

[Write your answer here]

9. Result and Conclusion

[Write your findings here]

10. Viva Questions

Q. 1 What is the electromagnetic spectrum?
Q. 2 What is the RF spectrum? What is its frequency range?
Q. 3 What is the difference between licensed and unlicensed spectrum?
Q. 4 What are ISM bands? Give examples.
Q. 5 Why is the 2.4 GHz band more congested than the 5 GHz band?
Q. 6 What is the role of the ITU in spectrum management?
Q. 7 What is the role of TRAI in spectrum management in India?
Q. 8 Why is spectrum considered a valuable national resource?
Q. 9 Explain the trade-off between frequency, range, and data rate.

11. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 1.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*, Chapter 1.
- **ITU** – Radio Regulations and Spectrum Allocation Charts.
- **TRAI** – Spectrum Auction and Allocation Documents.

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Experiment No. 02

Title:

Simulation of Basic Communication System (Tx–Channel–Rx) using PathView

1. Aim

To simulate a basic wireless communication system consisting of a transmitter, a wireless channel, and a receiver using the PathView online simulator.

Specific Objectives:

- To understand the functional blocks of a basic communication system (Transmitter, Channel, Receiver).
- To generate a sine wave signal at the transmitter.
- To simulate the effect of noise in the wireless channel by adding noise to the signal.
- To observe the received signal and analyze how noise affects signal quality.
- To study the relationship between noise level and signal distortion.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 1: Basics of Wireless Communication
Theory Section(s)	Elements of a Wireless Communication System
Course Outcome	CO1
Laboratory Outcome	LO1

3. Theoretical Background

3.1 Introduction

A wireless communication system transmits information from a source to a destination using electromagnetic waves as the transmission medium. The three fundamental building blocks of any communication system are the transmitter, the channel, and the receiver.

In a typical communication system, the information source generates a message (voice, data, or video). The transmitter processes this message into a form suitable for transmission over the channel. The channel is the physical medium through which the signal travels — in wireless systems, this is the atmosphere or free space. During transmission, the signal is corrupted by noise and interference. The receiver captures the signal from the channel, processes it, and attempts to recover the original information.

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3.2 Fundamental Concepts

a) Transmitter (Tx): The transmitter converts the information signal into a form suitable for transmission. In this experiment, we use a sine wave generator as the transmitter.

b) Wireless Channel: The channel is the medium through which the signal travels from the transmitter to the receiver. In wireless communication, the channel is the atmosphere. The transmitted signal is affected by various impairments, including:

- **Path Loss:** Signal strength decreases with distance.
- **Noise:** Unwanted random signals that corrupt the transmitted signal (e.g., thermal noise, atmospheric noise).
- **Interference:** Signals from other wireless systems operating on the same frequency.

In this experiment, we simulate the channel by adding **noise** to the transmitted signal.

c) Receiver (Rx): The receiver captures the signal from the channel and processes it. In this experiment, we use a Scope block to visualize the received signal.

3.3 Working Principle

The simulation follows a step-by-step process that mirrors the operation of a real communication system:

1. ***Signal Generation (Transmitter):*** A sine wave signal is generated.
2. ***Channel Simulation:*** Noise is added to the sine wave signal.
3. ***Signal Observation (Receiver):*** The noisy signal is displayed on a Scope.

3.4 Parameters to be Observed/Measured

- ***Signal Quality:*** How clear or distorted the received signal appears.
- ***Effect of Noise:*** The relationship between noise level and signal distortion.
- ***Transmitted vs. Received Signal:*** Comparison of the original signal and the noisy signal.

3.5 Practical Applications

- **Performance Evaluation:** This simulation is used to evaluate the performance of communication systems under different channel conditions.
- **System Design:** Engineers use such simulations to design and optimize communication systems before hardware implementation.
- **Educational Tool:** This simulation helps students understand the fundamental concepts of communication systems.

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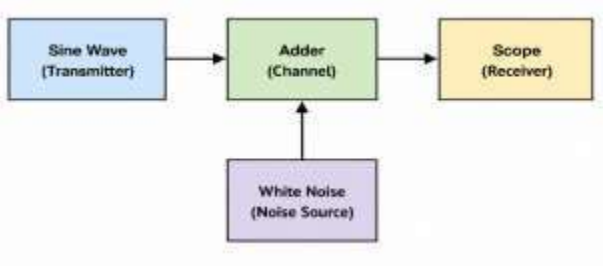
4. Requirements

Hardware: PC / Laptop with internet access

Software / Online Resources: Web browser to access PathView Simulator.

5. Experimental Setup

Since this is a software-based simulation experiment, the setup involves accessing the PathView website and building the simulation using drag-and-drop blocks.



6. Procedure

1. Open your browser, go to ***view.pathsim.org***, and press Enter.
2. Wait for the website to load completely. You will see a blank canvas (the workspace) with a sidebar on the left containing the Node Library marked as **'Blocks'**.
3. Important**:** No sign-up, login, or installation is required. The simulator runs entirely in your browser.
4. Wait for the website to load completely. You will see a blank canvas (the workspace) with a sidebar on the left containing the **Blocks** library.
5. **Add the Transmitter (Signal Source):**
    - Click on the **'Blocks'** panel on the left side of the screen.
    - Search for the **'SinusoidalSource'**. This block generates a sine wave signal.
    - Click on it. A small block 'SinusoidalSource' will appear on the canvas.
6. **Add the Noise Source (Channel Impairment):**
    - In the **'Blocks'** panel, search for the **'WhiteNoise'** block.
    - This block generates white noise, which we will use to simulate channel noise.
    - Click on it. A small block 'WhiteNoise' will appear on the canvas.
7. **Add the Adder (Combine Signal and Noise):**
    - In the **'Blocks'** panel, search for the **'Adder'** block.

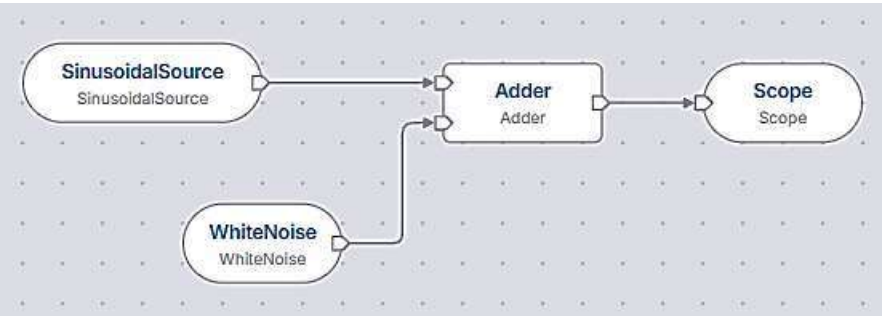
- This block adds two signals together.
- Click on it. A small block 'Adder' will appear on the canvas.

8. **Add the Receiver (Signal Visualizer):**
   - In the **'Blocks'** panel, search for the **'Scope'** block.
   - This block displays the signal as a waveform (like an oscilloscope).
   - Click on it. A small block 'Scope will appear on the canvas.

   Your canvas should now have four blocks: SinusoidalSource, WhiteNoise, Adder, and Scope.

9. **Connect the Transmitter to the Adder:**
   - Move your mouse over the SinusoidalSource block. You will see a small circle (port) on its right side. This is the output port. The mouse pointer (cursor) will change from hand icon to + (plus) sign.
   - Click and hold the output port of SinusoidalSource block.
   - Drag your mouse to the first input port of the Adder block (on its left side). A wire will appear as you drag.
   - Release the mouse button to complete the connection.
10. **Connect the Noise Source to the Adder:**
    - Click and hold on the output port (right side) of the WhiteNoise block.
    - Drag your mouse to the second input port of the Adder block (on its left side).
    - Release the mouse button to complete the connection.
11. **Connect the Adder to the Receiver:**
    - Click and hold on the output port (right side) of the Adder block.
    - Drag your mouse to the input port of the Scope block (on its left side).
    - Release the mouse button to complete the connection.

Your block diagram should now look like this:



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12. Look for a **'Run'** button in the top toolbar of the PathView interface. It may be a triangular "play" icon (▶).
13. Click the **'Run'** button. The simulation will start.
14. A plot window will open, showing the received signal (sine wave with noise).

Adjusting the Noise Level

15. Double click the WhiteNoise block on the canvas, a '**Properties'** panel will pop-up.
16. Find the standard_deviation parameter. This controls the strength (amplitude) of the noise.
    - Its default value is 1.0.
17. **To increase the noise:** Change the value to 2.0 or 5.0.
18. **To decrease the noise:** Change the value to 0.5 or 0.1.
19. **To remove noise entirely:** Change the value to 0.0.
20. After changing the value, click the **'Run'** button again to see the new result.

Observe how the waveform on the Scope changes as you increase or decrease the noise level.

Data Recording

21. For each noise level setting (standard_deviation), observe the signal on the Scope.
22. Record your observations in the table below.
23. Note how the signal quality changes as the noise level increases or decreases.

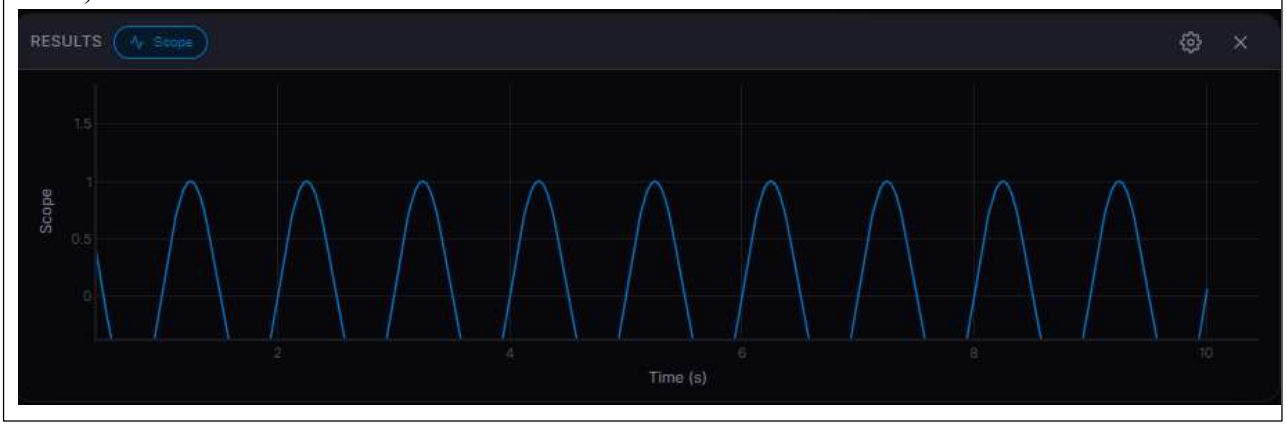
7. Observations

Table 1: Effect of Noise Level on Signal Quality

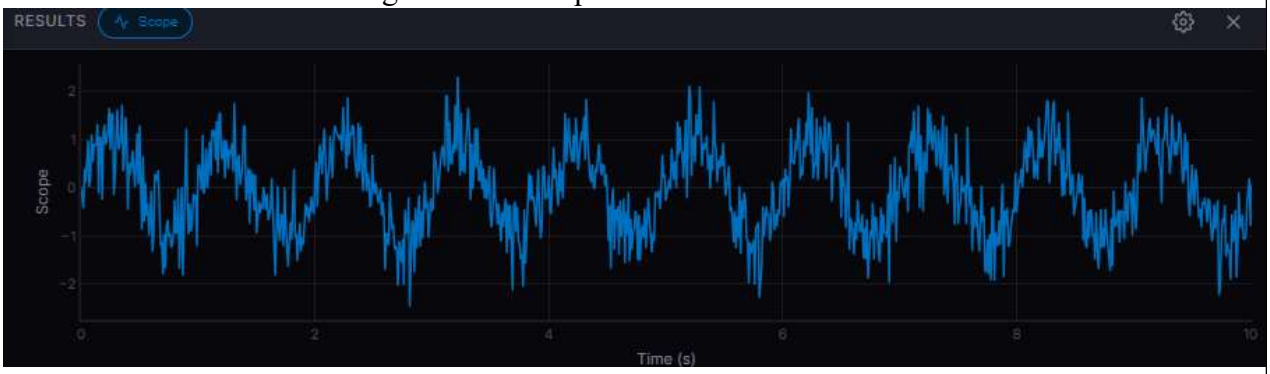
Noise Level (standard_deviation)	Signal Quality (Clean / Slightly Noisy / Very Noisy / Almost Lost)	Description of Observed Waveform
0.0	Clean	Perfect sine wave
0.1	Slight noise	Slight Fluctuation
0.5	Slight noise	Notable Noise
1.0	Noise	Distorted Signal
2.0	Loud noise	Signal barely visible
5.0	Lost almost	Signal Buried in noise

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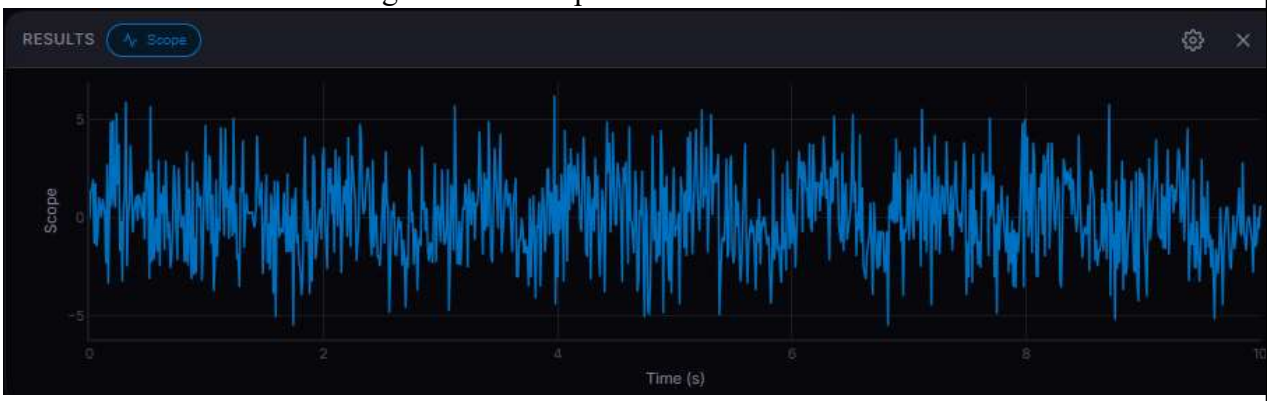
Paste the screenshots of the signal on the scope for Noise Level = 0 (i.e. transmitted signal without noise)



Paste the screenshots of the signal on the scope for Noise Level = 0.5



Paste the screenshots of the signal on the scope for Noise Level = 2.0.



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Table 2: Comparison of Transmitted and Received Signal

Parameter	Transmitted Signal (Sine)	Received Signal (Sine + Noise)
Waveform Shape	Smooth sine wave	Distorted sine wave with random fluctuations
Amplitude	Constant	Varies slightly due to noise
Noise Presence	No noise	Noise Present
Signal Quality	Perfect	Degraded

8. Analysis and Discussion

1. **From Table 1, observe the relationship between noise level and signal quality. What general trend do you observe?**

As the noise level increases, the signal quality decreases. At low noise levels, the sine wave remains clear and recognizable. As the noise becomes stronger, the waveform becomes more distorted, and at very high noise levels, the original signal is almost completely hidden by the noise.

2. **At very high noise level (e.g.,** standard_deviation = 5.0**), what do you observe about the received signal? Is it still recognizable as a sine wave?**

At **standard deviation = 5.0**, the received signal is dominated by noise and is **not easily recognizable as a sine wave**.

3. **At very low noise level (e.g.,** standard_deviation = 0.1**), what do you observe about the received signal? How does it compare to the original sine wave?**

At **standard_deviation = 0.1**, the received signal is **very similar to the original sine wave**, with only slight noise and minor distortion.

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4. **Why is it important to simulate a communication system before building it in hardware?**

Simulkation helps engineers test system performance,,identify problems reduce system cost and time .

9. Result and Conclusion

Noise has a significant effect on signal transmission. As the noise level increases, the received signal quality decreases. Lower noise produces a clearer and more accurate signal, while higher noise makes the signal difficult to detect.

10. Viva Questions

1. What are the three basic elements of a communication system?
2. What is the function of a transmitter?
3. What is the function of a receiver?
4. What is a wireless channel? What impairments does it introduce?
5. What is noise? How does it affect signal quality?
6. What is the effect of increasing noise on the received signal?
7. Why is a sine wave used as the transmitted signal in this experiment?
8. What is the role of the Scope block in this experiment?
9. How does this simulation relate to real wireless communication?
10. What is the advantage of using a simulator like PathView for this experiment?

11. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 1, Section 1.4: Elements of a Wireless Communication System.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*, Chapter 1.
- **PathView Documentation** – view.pathsim.org

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Experiment No. 03

Title:

Wi-Fi Network Scanning using Mobile Apps (SSID, RSSI, Channel Mapping)

1. Aim

To scan and analyze nearby Wi-Fi networks by measuring parameters such as SSID, RSSI, channel number, and frequency using the WiFi Analyzer (open-source) application on a smartphone.

Specific Objectives:

- To identify nearby Wi-Fi access points and their SSIDs.
- To measure the Received Signal Strength Indicator (RSSI) of different networks.
- To identify the channel and frequency band used by each network.
- To analyze channel utilization and identify congested channels.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 3: Wireless Networking Technologies
Theory Section(s)	Wi-Fi Technology (IEEE 802.11)
Course Outcome	CO3
Laboratory Outcome	LO2

3. Theoretical Background

3.1 Introduction

Wi-Fi (IEEE 802.11) is the most widely used wireless networking technology for providing internet and data connectivity. Wi-Fi networks operate in the unlicensed 2.4 GHz and 5 GHz frequency bands. Multiple Wi-Fi networks often coexist in the same environment, especially in residential areas, offices, and public spaces. These networks may interfere with each other, affecting performance and user experience.

Wi-Fi network scanning is a fundamental technique used to discover and analyze available wireless networks. It helps in understanding the wireless environment, identifying sources of interference, and optimizing network performance. Tools like WiFi Analyzer (open-source) by VREM Software Development provide a simple and effective way to perform such analysis using a smartphone.

3.2 Fundamental Concepts

a) **SSID (Service Set Identifier):** The SSID is the name of a Wi-Fi network. It is a case-sensitive string of up to 32 characters that identifies a particular 802.11 wireless LAN. Devices use the SSID to select the network they wish to join.

b) **BSSID (Basic Service Set Identifier):** The BSSID is the MAC address of the access point (AP) that is broadcasting the Wi-Fi network. It uniquely identifies the AP.

c) **RSSI (Received Signal Strength Indicator):** RSSI is a measure of the power present in a received radio signal. It is typically expressed in dBm (decibels relative to 1 milliwatt). RSSI values are negative, with values closer to 0 indicating a stronger signal.

RSSI Range (dBm)	Signal Quality	Description
-30 to -50	Excellent	Very strong signal; ideal for high data rates
-50 to -65	Good	Strong signal; reliable connection
-65 to -75	Fair	Moderate signal; may experience occasional slowdowns
-75 to -85	Poor	Weak signal; may experience frequent disconnections
Below -85	Very Poor	Unreliable connection; may not be usable

d) **Channel:** Wi-Fi networks operate on specific frequency channels. In the 2.4 GHz band, channels are spaced 5 MHz apart, but only channels 1, 6, and 11 are non-overlapping. In the 5 GHz band, there are many more non-overlapping channels available, reducing interference.

e) **Frequency Band:** Wi-Fi operates primarily in two frequency bands:

Band	Frequency Range	Characteristics
2.4 GHz	2.4 – 2.4835 GHz	Longer range, better penetration through walls, but more congestion and interference
5 GHz	5.15 – 5.875 GHz	Higher data rates, less interference, but shorter range and less wall penetration

f) Channel Congestion: When multiple Wi-Fi networks operate on the same or overlapping channels, they compete for the same spectrum, leading to reduced throughput, increased latency, and degraded performance. Channel congestion is a common problem in the 2.4 GHz band.

3.3 Working Principle

Wi-Fi analyzers work by passively listening to beacon frames broadcast by access points. Beacon frames contain information about the network, including the SSID, BSSID, supported data rates, and channel information. The application measures the received signal strength of these frames to calculate RSSI.

3.4 Parameters to be Observed/Measured

- ***SSID*:** Network name.
- ***BSSID*:** MAC address of the access point.
- ***RSSI*:** Received signal strength in dBm.
- ***Channel*:** Channel number used by the network.
- ***Frequency Band*:** 2.4 GHz or 5 GHz.
- ***Security*:** Security protocol used (e.g., WPA2, WPA3, Open).
- ***Channel Congestion*:** Number of networks on each channel.
- ***RSSI vs Distance*:** Variation of signal strength with distance.

3.5 Practical Applications

- ***Network Planning:*** Site surveys to determine optimal placement of access points.
- ***Interference Detection:*** Identifying sources of interference and selecting the least congested channel.
- ***Network Troubleshooting:*** Diagnosing poor Wi-Fi performance.
- ***Home Networking:*** Optimizing router placement and channel selection for better coverage.

4. Requirements

Hardware: Android smartphone with Wi-Fi capability and Wi-Fi router or access point

Software / Online Resources: WiFi Analyzer (open-source) by VREM Software Development

5. Experimental Setup

Since this is a practical hands-on experiment, the setup involves using a smartphone with the WiFi Analyzer app in an environment with multiple Wi-Fi networks.

6. Procedure

1. Install the **WiFi Analyzer (open-source)** app by VREM Software Development from the Google Play Store.
2. Open the app and grant the required permissions (location permission is typically required for scanning Wi-Fi networks).
3. Ensure Wi-Fi is enabled on your smartphone.
4. Familiarize yourself with the following views in the app:
    - **Network List View:** Shows all detected networks with SSID, RSSI, channel, frequency, and security.
    - **Channel Graph View:** Shows signal strength of networks across different channels.
    - **Time Graph View:** Shows signal strength variation of a selected network over time.
5. Stand at a fixed location (e.g., near your Wi-Fi router).
    - Record the following information for at least 5 visible Wi-Fi networks in Table 1: SSID, BSSID, RSSI (dBm), Channel, Frequency (GHz), Security
6. Switch to the **"Channel Graph"** view.
7. Observe which channels are most congested (many networks overlapping on the same channel).
8. Record your observations for both 2.4 GHz and 5 GHz bands.

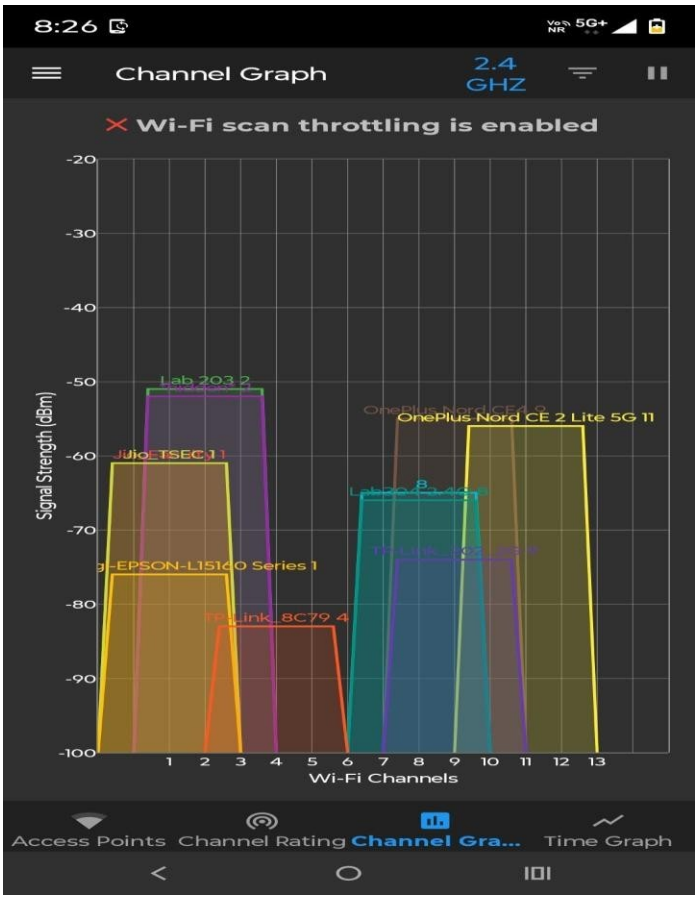
7. Observations

Screenshots: Paste screenshots here:

1. Network List View



2. Channel Graph View



3. Time Graph View (for your own network)

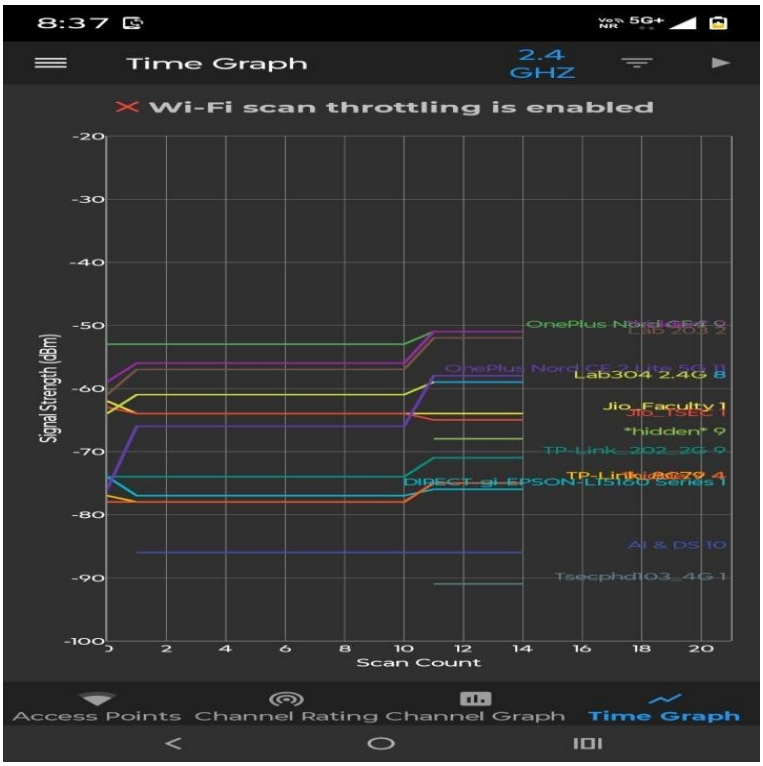


Table 1: Wi-Fi Network Parameters

Sr. No.	SSID	BSSID	RSSI (dBm)	Channel	Frequency (GHz)	Security
1	Lab 203	ac:15:a2:b9:a1:68	-51	2	2.417	[WPS WPA2]
2	*hidden*	ae:15:a2:a9:a1:68	-52	2	2.417	[WPA2]
3	OnePlus Nord CE4	c6:24:8d:55:3f:df	-55	9	2.452	[WPA2 WPA3]
4	OnePlus Nord CE 2 Lite 5G	de:ec:e2:3f:93:7e	-56	11	2.462	[WPA2 WPA3]

5	Lab 203	ac:15:a2:b9:a1:67	-57	36(42)	5.180	[WPS WPA2]
6	*hidden*	ae:15:a2:a9:a1:67	-59	36(42)	5.180	[WPA2]

Table 2: Channel Utilization Summary

Channel	Number of Networks	Channel Status
1	4	Congested
2	5	Congested
3	5	Congested
4	3	Congested
5	1	Free
6	2	Free
7	3	Congested
8	3	Congested
9	4	Congested
10	4	Congested
11	3	Congested

8. Analysis and Discussion

1. From Table 1, identify the network with the highest RSSI and the network with the lowest RSSI.

From Table 1:

- *Highest RSSI: Lab 203 — −51 dBm*
- *Lowest RSSI: hidden network — −59 dBm*

Since RSSI values closer to 0 dBm indicate a stronger signal, Lab 203 has the strongest signal, while the hidden network has the weakest.

2. From Table 2, which channel in the 2.4 GHz band is most congested? What are the consequences of channel congestion?

From Table 2, the most congested channels are Channel 2 and Channel 3, with 5 networks each.

Consequences of channel congestion:

- *Slower Wi-Fi speeds*
- *Increased interference*
- *Higher latency (delay)*
- *More packet loss and retransmissions*
- *Unstable or unreliable network connection*

Therefore, Channel 2 and Channel 3 are the most congested in the 2.4 GHz band.

9. Result and Conclusion

The Wi-Fi networks in the given area were successfully analyzed using RSSI, channel, frequency, and network count. Lab 203 had the highest RSSI of −51 dBm, indicating the strongest signal, while the hidden network had the lowest RSSI of −59 dBm. In the 2.4 GHz band, Channels 2 and 3 were the most congested, with 5 networks each. Channel 5 was identified as the best channel due to having only one network and a high channel rating. Channel congestion can cause interference, slower speeds, higher latency, and an unstable connection.

10. Viva Questions

1. What is SSID? How is it different from BSSID?
SSID (Service Set Identifier) is the name of a Wi-Fi network that users see when connecting to Wi-Fi.
BSSID (Basic Service Set Identifier) is the unique identifier of a specific Wi-Fi access point, usually represented by its MAC address.
Example:
SSID = Lab 203
BSSID = ac:15:a2:b9:a1:68

2. What is RSSI? How is it measured?
RSSI (Received Signal Strength Indicator) indicates the strength of a received Wi-Fi signal. It is usually measured in dBm (decibels relative to 1 milliwatt).
RSSI values are normally negative. A value closer to 0 dBm means a stronger signal.
Example: −40 dBm is stronger than −70 dBm**.**

3. What is the typical range of RSSI values for a good Wi-Fi signal?

A typical Wi-Fi signal can be classified approximately as:

RSSI	Signal Quality
−30 to −50 dBm	Excellent
−50 to −60 dBm	Good
−60 to −70 dBm	Fair
−70 to −80 dBm	Weak
Below −80 dBm	Very weak

Generally, **−30 to −60 dBm** is considered a good-to-excellent range.

4. What is the difference between the 2.4 GHz and 5 GHz frequency bands?

2.4 GHz	5 GHz
Longer range	Shorter range
Better penetration through walls	More affected by walls
Usually more congested	Usually less congested
Lower maximum speeds	Higher maximum
speeds More interference	Less interference

2.4 GHz is generally better for range, while **5 GHz** is better for speed and lower interference.

5. What is channel congestion? What problems does it cause?

Channel congestion occurs when many Wi-Fi networks use the same or overlapping channels in an area.

It can cause:

- Reduced Wi-Fi speed
- Increased interference
- Higher latency
- Packet loss and retransmissions
- Unstable connections

6. How can a Wi-Fi analyzer help in optimizing a wireless network?

A Wi-Fi analyzer scans nearby wireless networks and provides information such as SSID, BSSID, RSSI, channel, and frequency.

It can help to:

- Find less congested channels
- Identify sources of interference
- Check signal strength
- Choose the best Wi-Fi channel
- Improve network speed and reliability

7. What is the role of a beacon frame in Wi-Fi networks?

A beacon frame is a management frame periodically transmitted by a Wi-Fi access point. It announces the presence and characteristics of the network.

It contains information such as the SSID, supported data rates, channel information, and security capabilities. This allows nearby devices to discover and identify the Wi-Fi network.

11. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 3, Section 3.2: Wi-Fi Technology.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*.
- **Stallings, W.** – *Data and Computer Communications*.

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Experiment No. 04

Title:

Measurement of Signal Strength vs Distance for Wi-Fi Router

1. Aim

To measure the Received Signal Strength Indicator (RSSI) of a Wi-Fi network at increasing distances from the router, and to analyze the relationship between signal strength and distance to understand the concept of path loss.

Specific Objectives:

- To measure RSSI (in dBm) at various distances (in steps) from the Wi-Fi router.
- To observe the decrease in signal strength with increasing distance.
- To record and tabulate the measured values for analysis.
- To relate the observed trend to the concept of path loss in wireless communication.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 4: Signal Propagation and Interference
Theory Section(s)	4.4 Path Loss and Shadowing
Course Outcome	CO4
Laboratory Outcome	LO3

3. Theoretical Background

3.1 Introduction

In wireless communication, the signal strength of a Wi-Fi network decreases as the distance between the transmitter (router) and the receiver (smartphone) increases. This reduction in signal power is known as path loss. Path loss is a natural phenomenon that occurs because the transmitted energy spreads over a larger area as it travels away from the source. Understanding this relationship is fundamental to network planning, coverage analysis, and interference management.

This experiment provides a practical demonstration of path loss by measuring the Received Signal Strength Indicator (RSSI) of a Wi-Fi router at various distances using a smartphone with a Wi-Fi analyzer app. The measured values will be analyzed to understand the trend.

3.2 Fundamental Concepts

a) Path Loss: Path loss is the reduction in signal power as an electromagnetic wave travels from the transmitter to the receiver. It is a natural phenomenon and is present in every wireless communication system. The primary factor affecting path loss is distance; as distance increases, the signal spreads over a larger area, reducing the power density at the receiver.

b) RSSI (Received Signal Strength Indicator): RSSI is a measure of the power present in a received radio signal. It is typically expressed in dBm (decibels relative to 1 milliwatt). RSSI values are negative, with values closer to 0 indicating a stronger signal.

c) Inverse Square Law (Conceptual): The signal strength decreases proportionally to the square of the distance from the transmitter. In indoor environments, signal degradation is often faster due to reflections, absorption, and scattering by walls and objects.

3.3 Working Principle

When a smartphone connects to a Wi-Fi network, the Wi-Fi analyzer app displays the RSSI of the received signal. By moving the smartphone away from the router in measured steps and recording the RSSI at each location, the relationship between distance and signal strength can be observed.

3.4 Parameters to be Observed/Measured

- **Distance:** The distance from the Wi-Fi router (measured in steps).
- **RSSI:** The received signal strength (in dBm) as displayed by the app.
- **Signal Quality:** A qualitative assessment (Excellent/Good/Fair/Poor) based on the RSSI value.

3.5 Practical Applications

- **Network Planning:** Site surveys to determine the optimal placement of access points.
- **Coverage Mapping:** Identifying areas with weak signals (dead zones).
- **Troubleshooting:** Diagnosing poor Wi-Fi performance in specific locations.
- **Interference Analysis:** Understanding how obstacles and distance affect signal quality.

4. Requirements

Hardware: Android smartphone with Wi-Fi capability and Wi-Fi router or access point

Software / Mobile Application: WiFi Analyzer (open-source) by VREM Software Development

5. Experimental Setup

This experiment requires a clear path (line-of-sight) from the Wi-Fi router up to a distance of approximately 40-50 steps. The setup involves using a smartphone with the Wi-Fi analyzer app to measure RSSI at various distances along a straight path.

6. Procedure

1. Place the Wi-Fi router at a fixed location with no immediate obstacles blocking the signal.
2. Ensure the smartphone is connected to the Wi-Fi network of the router.
3. Open the WiFi Analyzer app on the smartphone and ensure it is displaying the RSSI of the network you wish to measure.
4. Stand near the router (0 steps) and note the RSSI reading in the app. Record it in the observation table.
5. Walk 5 steps away from the router, stop, wait for the RSSI reading to stabilize, and record the value.
6. Repeat the measurement at intervals of 5 steps up to 30 steps.
7. Ensure that the line-of-sight between the smartphone and the router is as clear as possible at each measurement point.
8. Fill in the Table 1 with the measured RSSI values at each distance.
9. Note any observations about the signal quality or environmental factors (e.g., obstacles, interference).

Note: You can measure the length of your step once using a measuring tape to get an approximate distance in meters if required.

Number of Steps	Approximate Distance (m)
0 steps	0 m
5 steps	~3.75 m
10 steps	~7.5 m

15 steps	~11.25 m
20 steps	~15 m
25 steps	~18.75 m
30 steps	~22.5 m
(Assumption: 1 step ≈ 0.75 meters)	

7. Observations

Table 1: RSSI vs Distance

Distance (Steps)	Approximate Distance (m)	RSSI (dBm)	Signal Quality (Excellent/Good/Fair/Poor)
0 steps	0 m	-39	Excellent
5 steps	~3.75 m	-45	Good
10 steps	~7.5 m	-45	Good
15 steps	~11.25 m	-50	Fair
20 steps	~15 m	-51	Fair
25 steps	~18.75 m	-56	Fair
30 steps	~22.5 m	-66	Poor

8. Analysis and Discussion

11. From Table 1, what is the general trend in RSSI values as distance from the router increases? Explain why this happens.

As the distance from the router increases, the RSSI value generally decreases (becomes more negative). This means the Wi-Fi signal becomes weaker as the smartphone moves farther away from the router. This happens because the wireless signal loses energy as it travels over a greater distance.

2. At what distance (in steps) does the signal quality become "Fair" or "Poor"? What are the practical implications of this for Wi-Fi coverage?

The signal quality becomes Fair or Poor at the distance where the RSSI falls below approximately −60 to −70 dBm, according to the values in Table 1.

This means that at greater distances, users may experience slower speeds, higher latency, and less reliable connections. Therefore, the router should be placed appropriately to provide good coverage throughout the required area.

3. **If you were to place obstacles (e.g., walls, furniture) between the router and the smartphone, how would the RSSI readings change?**

   Obstacles such as **walls and furniture would generally reduce the RSSI**, making the value more negative. This happens because obstacles can absorb, reflect, or weaken the Wi-Fi signal as it travels between the router and smartphone. As a result, the connection may become weaker and less reliable.

9. Result and Conclusion

The experiment showed that RSSI decreases as the distance between the router and smartphone increases, indicating that the Wi-Fi signal becomes weaker with increasing distance. The signal quality changes from Good to Fair or Poor at greater distances. Obstacles such as walls and furniture can further reduce the RSSI by weakening the wireless signal. Therefore, proper router placement and minimizing obstacles are important for maintaining good Wi-Fi coverage and a reliable connection.

10. Viva Questions

1. **What is RSSI? How is it measured?**
   RSSI (Received Signal Strength Indicator) indicates the strength of a received Wi-Fi signal. It is usually measured in dBm (decibels relative to one milliwatt). A value closer to 0 dBm represents a stronger signal.
2. **What is path loss? What are its primary causes?**
   Path loss is the reduction in signal strength as a wireless signal travels from the transmitter to the receiver. Its main causes include distance, obstacles, absorption, reflection, diffraction, and interference.
3. **Why does signal strength decrease with distance?**
   Signal strength decreases with distance because the wireless signal spreads out as it travels away from the transmitter, causing the signal energy to become weaker.
4. **What is the inverse square law? How does it relate to wireless signal propagation?**
   The inverse square law states that the power of a signal decreases approximately with the

square of the distance from the source. Therefore, as the distance increases, the received wireless signal becomes weaker.

5. **What factors (other than distance) affect Wi-Fi signal strength?**
   Factors include walls and other obstacles, interference from other wireless networks, building materials, antenna position, frequency band, and other electronic devices.
6. **How can the results of this experiment be used in real-world network planning?**
   The results can help determine suitable router locations, estimate Wi-Fi coverage areas, identify weak-signal zones, and decide where additional access points may be needed.
7. **Why is it important to take readings at fixed intervals?**
   Taking readings at fixed intervals ensures consistent measurements and makes it easier to compare how RSSI changes as the distance increases.
8. **Give examples of obstacles that can affect Wi-Fi signal strength.**
   Examples include concrete walls, brick walls, metal objects, glass, furniture, doors, and other large physical objects.

11. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 4, Section 4.4: Path Loss and Shadowing.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*.
- **Stallings, W.** – *Data and Computer Communications*.

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Experiment No. 05

Title:

Analysis of Channel Interference in the 2.4 GHz Band

1. Aim

To analyze the distribution of Wi-Fi networks over different channels in the 2.4 GHz band and identify possible channel interference using the WiFi Analyzer (open-source) mobile application.

Specific Objectives:

- To visualize the distribution of Wi-Fi networks across the 2.4 GHz frequency band.
- To identify channels with maximum and minimum network occupancy.
- To observe Co-Channel Interference (CCI) and Adjacent Channel Interference (ACI).
- To understand the concept of non-overlapping channels (1, 6, and 11).
- To recommend the best channel for deploying a new Wi-Fi router based on observed interference.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 4: Signal Propagation and Interference
Theory Section(s)	4.7 Interference in Wireless Systems
Course Outcome	CO4
Laboratory Outcome	LO3

3. Theoretical Background

3.1 Introduction

In wireless communication, interference occurs when unwanted signals from other sources degrade the quality of the desired signal. In Wi-Fi networks, interference is a common problem, especially in the 2.4 GHz band where many devices operate.

Wi-Fi networks operate in the unlicensed 2.4 GHz and 5 GHz frequency bands. The 2.4 GHz band, while offering longer range and better wall penetration, is crowded because many devices — Wi-Fi routers, Bluetooth devices, and cordless phones — share the same spectrum. This leads to congestion, interference, and degraded performance.

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This experiment uses the WiFi Analyzer app to visualize the 2.4 GHz spectrum and observe channel interference in real-world conditions.

3.2 Fundamental Concepts

a) Channel: A Wi-Fi channel is a specific frequency range within a frequency band. In the 2.4 GHz band, the IEEE 802.11 standard defines up to 14 channels, each spaced 5 MHz apart. The number of usable channels depends on the country's regulatory domain. In India, channels 1 to 13 are available.

Channel	Center Frequency (GHz)	Frequency Range (GHz)
1	2.412	2.401 – 2.423
2	2.417	2.406 – 2.428
3	2.422	2.411 – 2.433
4	2.427	2.416 – 2.438
5	2.432	2.421 – 2.443
6	2.437	2.426 – 2.448
7	2.442	2.431 – 2.453
8	2.447	2.436 – 2.458
9	2.452	2.441 – 2.463
10	2.457	2.446 – 2.468
11	2.462	2.451 – 2.473
12	2.467	2.456 – 2.478
13	2.472	2.461 – 2.483

b) Channel Width and Non-Overlapping Channels: A standard Wi-Fi signal occupies 20-22 MHz of bandwidth. Since channels are spaced only 5 MHz apart, adjacent channels overlap significantly. Only channels separated by at least 25 MHz (i.e., 5 channels apart) do not overlap. Channels 1, 6, and 11 are the only non-overlapping channels in the 2.4 GHz band.

c) Co-Channel Interference (CCI): CCI occurs when two or more networks operate on the same channel in the same area. Their signals compete for the same spectrum, leading to reduced throughput, increased latency, and degraded performance.

d) Adjacent Channel Interference (ACI): ACI occurs when networks operate on neighboring channels whose signals overlap. For example, Channel 1 and Channel 2 overlap because their frequency ranges intersect.

e) RSSI (Received Signal Strength Indicator): RSSI is a measure of the power present in a received radio signal. It is expressed in dBm (decibels relative to 1 milliwatt). RSSI values are negative, with values closer to 0 indicating a stronger signal.

RSSI Range (dBm)	Signal Quality
-30 to -50	Excellent

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-50 to -65	Good
-65 to -75	Fair
-75 to -85	Poor
Below -85	Very Poor

3.3 Working Principle

WiFi Analyzer works by passively listening to beacon frames broadcast by access points. Each beacon frame contains information about the network, including its SSID, BSSID (MAC address), channel, and signal strength. The app aggregates this data and displays it graphically.

The Channel Graph view shows the signal strength of all detected networks across the frequency spectrum. Each network appears as a curve (parabola) centered on its operating channel. The height of the curve indicates signal strength (RSSI), and the width indicates signal spread across adjacent frequencies.

3.4 Parameters to be Observed/Measured

- ***Channel:*** The channel number on which each network operates.
- ***RSSI***: The received signal strength of each network (in dBm).
- ***Channel Occupancy:*** The number of networks on each channel.
- ***Co-Channel Interference:*** Multiple networks on the same channel.
- ***Adjacent Channel Interference:*** Networks on overlapping adjacent channels.

3.5 Practical Applications

- ***Wi-Fi Planning:*** Selecting the least congested channel for new router deployment.
- ***Interference Detection:*** Identifying sources of interference in the 2.4 GHz band.
- ***Network Troubleshooting:*** Diagnosing poor Wi-Fi performance due to channel congestion.
- ***Site Survey:*** Understanding the wireless environment before deploying access points.

4. Requirements

Hardware: Android smartphone with Wi-Fi capability

Software / Mobile Application: WiFi Analyzer (open-source) by VREM Software Development

5. Experimental Setup

This experiment requires a location with multiple Wi-Fi networks. The setup involves using a smartphone with the WiFi Analyzer app to observe the 2.4 GHz channel graph.

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6. Procedure

1. Select a location with multiple active Wi-Fi networks (e.g., college campus, apartment building, hostel, office, shopping mall).
2. Enable Wi-Fi and Location Services on your smartphone.
3. Open the WiFi Analyzer (open-source) app.
4. Wait approximately 20 seconds for the scan to stabilize.
5. Ensure the app is set to scan the 2.4 GHz band (look for a toggle or tab to switch between 2.4 GHz and 5 GHz).
6. Navigate to the 'Channel Graph' view (usually at the bottom of the app).
7. Observe the X-axis — it represents Wi-Fi channels (1 to 13).
8. Observe the Y-axis — it represents Signal Strength (RSSI in dBm).
9. Each parabolic curve represents one Wi-Fi network, with its peak at its center channel. The height of the curve indicates signal strength (RSSI), and the width indicates signal spread across adjacent frequencies.
10. Identify the visible networks on the Channel Graph. For each network, record its SSID, channel, and RSSI in Table 1.
11. Note the number of networks on each channel in Table 2.
12. ***Observe Co-Channel Interference (CCI):*** Look for multiple networks whose curves have their peaks on the same channel. These networks interfere with each other.
13. ***Observe Adjacent Channel Interference (ACI):*** Look for networks on neighboring channels whose curves overlap. This indicates ACI. Record your observations in Table 3.
14. Identify the channel(s) that are free or least congested.
15. Recommend the best channel for deploying a new Wi-Fi router based on your observations.

7. Observations

Table 1: Signal Strength vs Channel

Network	Channel	RSSI (dBm)	Signal Quality
OnePlus Nord CE4	1	-51 dBm	Excellent
Lab 203	2	-54 dBm	Excellent
Jio_TSEC	1	-57 dBm	Good

hidden	2	-60 dBm	Good
Jio_Faculty	1	-61 dBm	Good

Table 2: Channel Utilization Summary

Channel	Number of Networks	Network Names	Channel Status
1	4	OnePlus Nord CE4, Jio_TSEC, Jio_Faculty, DIRECT-gi-EPSON-L15160 Series	Highly Congested
2	2	Lab 203, *hidden*	Moderately Utilized
3	1	Lab304 2.4G	Lightly Utilized
4	1	Ph.D Lab 103	Lightly Utilized
5	0	None	Free
6	1	Tsec... 4G	Lightly Utilized
7	1	TP-Link_302	Lightly Utilized
8	0	None	Free
9	2	*hidden*, TP Link 202_2G	Moderately Utilized
10	2	AI & DS, *hidden*	Moderately Utilized
11	1	Redmi 9 Prime	Lightly Utilized
12	0	None	Free

Table 3: Co-Channel and Adjacent Channel Interference Observation

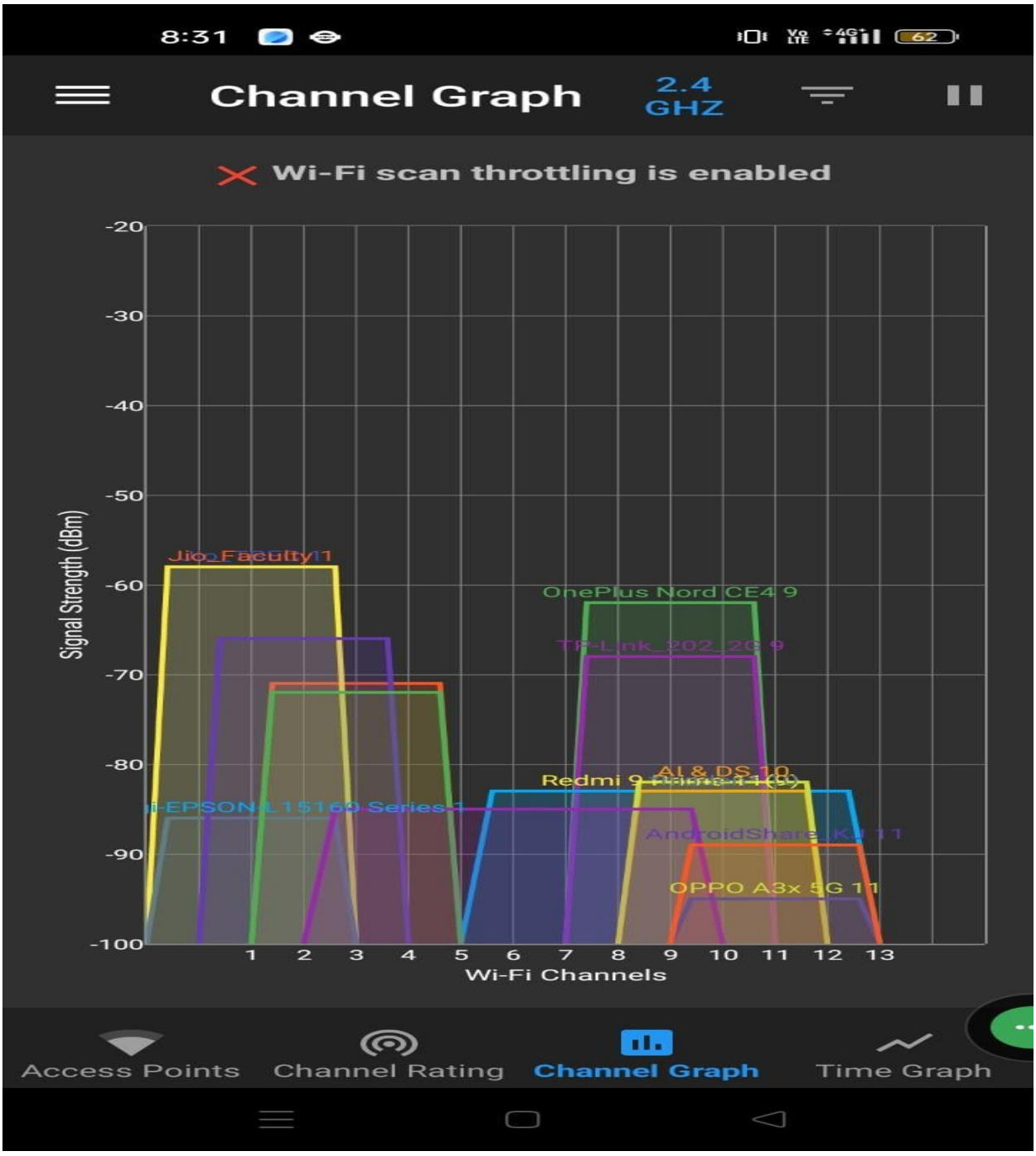
Observation Type	Channel(s) Affected	Networks Involved	Remarks
Co-Channel Interference (CCI)	1, 2, 9, 10	**CH 1:** OnePlus, Jio_TSEC, Jio_Faculty, EPSON	Significant CCI on Channel 1 due to four distinct networks

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		CH 2: Lab 203, *hidden* **CH 9:** *hidden*, TP Link 202_2G	competing for the same exact frequency space.
Adjacent Channel Interference (ACI)	1, 2, 3, 4, 9, 10, 11	All networks spanning Channels 1 through 4, and 9 through 11.	High ACI observed because most networks are grouped on overlapping channels instead of the standard non-overlapping 2.4 GHz channels (1, 6, 11). For instance, Channels 1, 2, 3, and 4 heavily overlap each other's 20 MHz bands.

Screenshots

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8. Analysis and Discussion

1. **Which channel in the 2.4 GHz band is most congested?**

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Channel 1 has the highest number of access points

2. **How did you identify Co-Channel Interference (CCI) on the Channel Graph? Which channels showed maximum CCI?**

   To identify cci you should see for channels that have the same peak, channel 1 showed the maximum one

3. **How did you identify Adjacent Channel Interference (ACI) on the Channel Graph? Which overlapping channels did you observe?**

   Look for networks on neighboring channels whose curves overlap, channel 1,2,3 and 4 show the highest

4. **Which channel would you recommend for configuring a new Wi-Fi router at your observation location? Justify your answer.**

   Channel 5,8 and 11 have the lowest interference so if we want to setup any wifi network we can do it here

9. Result and Conclusion

The surveyed area exhibits significant variations in Wi-Fi signal strength. A few primary access points, particularly “OnePlus Nord CE4” (-51 dBm) on Channel 1 and “Lab 203” (-49 to -54 dBm) on Channel 2, provide excellent signal quality. However, the signal strength of several other networks decreases to a fair level as RSSI falls below -65 dBm.

Channel utilization is also a major concern. Channel 1 is heavily congested, with four distinct access points operating on the same channel. This increases the likelihood of co-channel contention, packet collisions, and reduced network performance for connected devices.

The 2.4 GHz channel allocation is poorly optimized, with many networks operating on overlapping channels such as 2, 3, 4, 7, 9, and 10, rather than the recommended non-overlapping channels 1, 6, and 11. This inefficient allocation significantly increases interference within the wireless environment.

Co-Channel and Adjacent-Channel Interference

- Co-Channel Interference (CCI) is particularly noticeable on Channels 1, 2, 9, and 10, where multiple access points share the same frequency and must compete for airtime.

- Adjacent Channel Interference (ACI) is also severe due to the extensive use of overlapping channels. For example, the strong "Lab 203" network on Channel 2 significantly overlaps with networks operating on Channels 1, 3, and 4. This increases background noise and reduces the effective signal-to-noise ratio (SNR).

Overall, the current 2.4 GHz Wi-Fi environment is highly congested and poorly optimized. The concentration of access points across overlapping lower channels (1–4) and upper channels (9–11) creates substantial CCI and ACI. Consequently, users may experience reduced throughput, increased latency, packet loss, and connection instability, even when the received signal strength appears strong.

10. Viva Questions

1. What is a Wi-Fi channel?

A Wi-Fi channel is a smaller portion of a wireless frequency band, such as **2.4 GHz or 5 GHz**, that a router uses to transmit and receive data. It can be compared to a **lane on a highway**, where data travels between the router and connected devices.

2. How many channels are available in the 2.4 GHz band in India?

In India, 13 Wi-Fi channels (Channels 1–13) are available for use in the 2.4 GHz frequency band, subject to applicable regulations.

3. What is Co-Channel Interference (CCI)?

Co-Channel Interference (CCI) occurs when two or more Wi-Fi networks operate on the same channel. Since Wi-Fi uses a *listen-before-talk* mechanism, devices sharing the channel must wait for their turn to transmit. Although this helps avoid data collisions, excessive channel sharing can reduce throughput and increase network latency.

4. What is Adjacent Channel Interference (ACI)?

Adjacent Channel Interference (ACI) occurs when Wi-Fi networks operate on overlapping channels, such as Channels 1 and 2. Because their signals partially overlap, transmissions can interfere with one another, increasing noise and reducing communication quality. This can result in packet loss, retransmissions, reduced speeds, and unstable connections. In many situations, ACI can have a more significant impact than CCI because overlapping transmissions are not coordinated as effectively.

5. Why are Channels 1, 6, and 11 considered non-overlapping?

In the 2.4 GHz Wi-Fi band, channels are spaced only 5 MHz apart, while a typical Wi-Fi transmission occupies approximately 20 MHz of bandwidth. Channels 1, 6, and 11 are sufficiently

separated so that their standard 20 MHz signal bandwidths do not overlap. Therefore, using these three channels helps minimize Adjacent Channel Interference (ACI).

6. How does channel congestion affect Wi-Fi performance?

When multiple networks and devices compete for the same wireless spectrum, congestion can negatively affect network performance. Common effects include:

- Slower speeds: Devices have less available airtime for transmitting data.
- Higher latency: Devices must wait longer for an opportunity to transmit.
- Packet loss and connection drops: Increased interference and noise can cause packets to be corrupted or lost, resulting in retransmissions and occasional disconnections.

7. Which channel would you recommend for a new Wi-Fi router in a congested area?

For a 2.4 GHz network, it is generally recommended to use Channel 1, 6, or 11. A Wi-Fi analyzer can be used to determine which of these channels has the least interference and fewest strong neighboring networks, allowing you to select the most suitable option.

If the 2.4 GHz band is heavily congested, switching compatible devices to the 5 GHz band is often preferable because it provides more channel options and generally experiences less interference.

8. How can a Wi-Fi analyzer help in optimizing a wireless network?

A Wi-Fi analyzer provides a visual representation of the surrounding radio-frequency (RF) environment, making it easier to identify and resolve wireless issues. It can help you:

- Identify neighboring networks and the channels they are using.
- Detect CCI and ACI affecting network performance.
- Measure RSSI/signal strength in different areas of a building.
- Locate Wi-Fi dead zones where signal coverage is weak.
- Select an appropriate channel based on actual interference and signal conditions.
- Make informed network configuration decisions to improve coverage, reliability, and performance.

11. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 4, Section 4.7: Interference in Wireless Systems.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*.
- **Stallings, W.** – *Data and Computer Communications*.

Experiment No. 06

Title:

Study of Path Loss and Shadowing in an Indoor Environment

1. Aim

To study the effect of different indoor obstacles on Wi-Fi signal strength by measuring the Received Signal Strength Indicator (RSSI).

Specific Objectives:

- Observe the effect of indoor obstacles on wireless signal propagation.
- Measure Wi-Fi signal strength using RSSI values.
- Compare the attenuation caused by different obstacles.
- Understand the effect of obstacles on indoor wireless coverage.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 4: Signal Propagation and Interference
Theory Section(s)	4.4 Pathloss and Shadowing
Course Outcome	CO4
Laboratory Outcome	LO5

3. Theoretical Background

3.1 Introduction

In wireless communication, the transmitted signal becomes weaker as it travels from the transmitter to the receiver. This reduction in signal strength is known as path loss.

In an indoor environment, the wireless signal may encounter obstacles such as walls, doors, glass partitions, and furniture. These obstacles can absorb, reflect, scatter, or partially block the signal. As a result, the signal received by the wireless device may become weaker.

The additional reduction or variation in received signal strength caused by large obstacles in the propagation path is known as shadowing.

Therefore, indoor wireless coverage depends not only on the distance between the transmitter and receiver but also on the type, thickness, and number of obstacles present in the signal path.

In this experiment, the approximate distance between the Wi-Fi router and smartphone is kept constant. The RSSI is then measured under different obstacle conditions. Thus, the experiment focuses on the effect of obstacles on indoor wireless signal propagation.

3.2 RSSI

RSSI stands for Received Signal Strength Indicator. It represents the strength of the wireless signal received by a device and is generally displayed in dBm. Since RSSI values are negative, a value closer to zero represents a stronger received signal.

RSSI Range (dBm)	Signal Quality
-30 to -50	Excellent
-50 to -65	Good
-65 to -75	Fair
-75 to -85	Poor
Below -85	Very Poor

3.3 Effect of Indoor Obstacles

Different obstacles affect wireless signals differently. When a wireless signal encounters an obstacle, part of its energy may be absorbed, reflected, scattered, or blocked.

A wooden door or glass partition may produce a different amount of attenuation compared with a thick brick or concrete wall. Similarly, the presence of multiple obstacles can further reduce the received signal strength.

The observations obtained in this experiment are useful for understanding Wi-Fi coverage inside buildings and for deciding suitable locations for Wi-Fi Access Points.

4. Requirements

1. Wi-Fi Router / Access Point.
2. Android smartphone with Wi-Fi capability.
3. WiFi Analyzer application.
4. Indoor environment containing different obstacles such as doors, partitions, or walls.

5. Procedure

Step 1: Wi-Fi Setup

- Switch ON the Wi-Fi router.
- Open the WiFi Analyzer application and identify the Wi-Fi network to be tested.

Step 2: LOS Reference Measurement

- Select a measurement position with a clear Line-of-Sight (LOS) path between the router and smartphone. Keep the distance approximately fixed, preferably around 3–4 m or any suitable distance available in the laboratory.
- Observe the RSSI for approximately 20–30 seconds and record a representative stable value. This will be considered the baseline RSSI.

Step 3: Measurement with Obstacles

- Select measurement positions at approximately the same distance from the router, but with different obstacles in the signal path.
  - ***Possible obstacle conditions include:*** ○ Wooden door. ○ Glass partition. ○ Brick or concrete wall.
  - Multiple walls or obstacles, if available.
- For each condition, observe the RSSI for approximately 20–30 seconds and record the representative stable value.

Step 4: Compare the Results

- Compare the RSSI obtained under different obstacle conditions with the LOS baseline.
- Determine which obstacle causes the maximum reduction in received signal strength.

6. Observations

Table 1: Effect of Indoor Obstacles on Wi-Fi Signal Strength

Sr. No.	Obstacle Condition	Approx. Distance from Router (m)	RSSI (dBm)	Signal Reduction from LOS Baseline (dB)
1	Clear LOS / No Obstacle	3.4	−51	0
2	Wooden Door	4.9	−54	3 dB
3	Glass Partition	8.7	−59	8 dB
4	Brick / Concrete Wall	24.8	−68	17 dB
5	Multiple Walls / Obstacles	48.8	−74	23 dB

Note:

Signal Reduction from LOS Baseline = LOS Baseline RSSI − RSSI under Obstacle Condition

For example, if:

- LOS Baseline RSSI = −50 dBm

- RSSI with obstacle = −65 dBm

then:

Signal Reduction = −50 − (−65) = 15 dB

7. Analysis and Discussion

1. Arrange the tested obstacle conditions from least attenuation to maximum attenuation.

Walls/Obstacles Clear LOS → Wooden Door → Glass Partition → Brick/Concrete Wall → Multiple.

2. Compare the signal reduction caused by a single obstacle and multiple obstacles.

A single obstacle causes less signal reduction compared to multiple obstacles. In the observed readings, the wooden door caused a 3 dB reduction, while multiple walls/obstacles caused a 23 dB reduction.

3. Which obstacle would require greater consideration while planning the location of a WiFi Access Point? Give a reason based on your observations.

Brick/concrete walls and multiple obstacles require greater consideration because they cause higher signal reduction. Therefore, the Access Point should be placed where fewer obstacles are present between the transmitter and receiver.

4. How can the results of this experiment be used to identify possible weak coverage areas or dead zones inside a building?

RSSI measurements can be taken at different locations inside the building. Areas with very low RSSI indicate weak coverage or possible Wi-Fi dead zones. These areas can then be improved by changing the Access Point location or adding additional Access Points.

5. Based on your observations, suggest one practical method to improve Wi-Fi coverage in an area having significant signal reduction due to obstacles.

A practical method is to reposition the Wi-Fi Access Point to reduce the number of walls and other obstacles between the Access Point and the user. Additional Access Points can also be installed if required.

8. Result and Conclusion

The effect of different indoor obstacles on Wi-Fi signal strength was studied using RSSI measurements. The LOS condition showed the strongest signal of −51 dBm, while the multiple walls/obstacles condition showed the weakest signal of −74 dBm. The maximum signal reduction

observed was 23 dB. Hence, indoor obstacles significantly affect wireless signal propagation and Wi-Fi coverage.

9. Viva Questions

1. What is shadowing in wireless communication?

Shadowing is the additional reduction or variation in received signal strength caused by large obstacles in the propagation path.

2. What is the difference between LOS and NLOS propagation?

LOS propagation has a clear direct path between the transmitter and receiver, whereas NLOS propagation has obstacles blocking the direct path.

3. How do physical obstacles affect wireless signal propagation?

Obstacles can absorb, reflect, scatter or partially block the wireless signal, resulting in reduced received signal strength.

4. What is meant by obstacle-induced attenuation?

It is the reduction in wireless signal strength caused by an obstacle present between the transmitter and receiver.

5. Why may different materials produce different amounts of signal reduction?

Different materials have different properties and therefore absorb, reflect and transmit wireless signals differently.

6. What happens to wireless coverage when multiple obstacles are present in the signal path? Multiple obstacles generally cause greater signal attenuation and reduce wireless coverage.

7. What is a Wi-Fi dead zone?

A Wi-Fi dead zone is an area where the wireless signal is too weak or unreliable for proper connectivity.

8. How can the placement of an Access Point affect indoor Wi-Fi coverage?

Proper placement of an Access Point can reduce obstacles in the signal path and improve indoor Wi-Fi coverage.

9. Why is an LOS condition used as a reference in this experiment?

LOS provides a relatively unobstructed reference RSSI against which the effect of different obstacles can be compared.

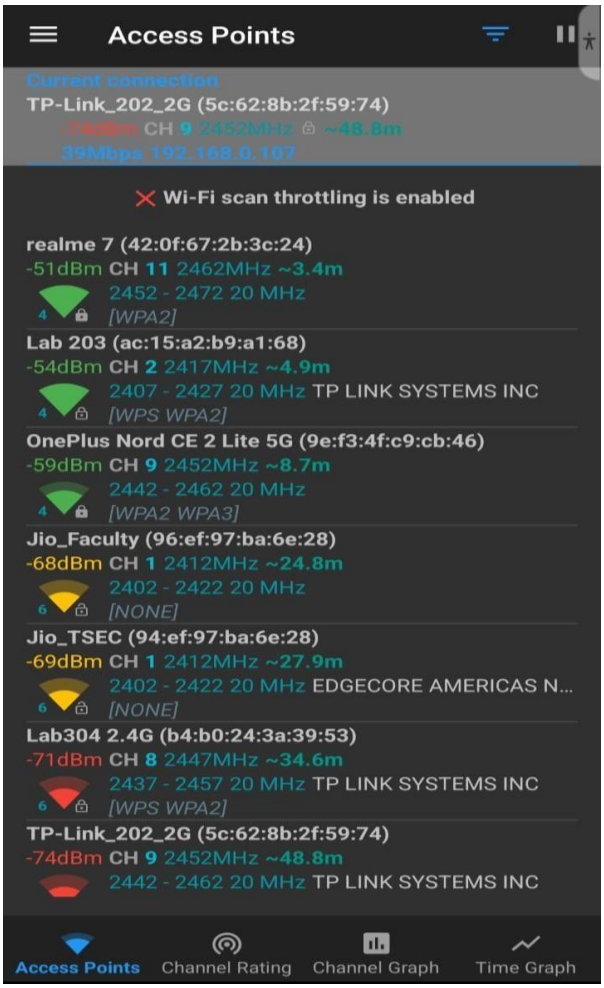
10. How can additional Access Points help improve indoor wireless coverage?

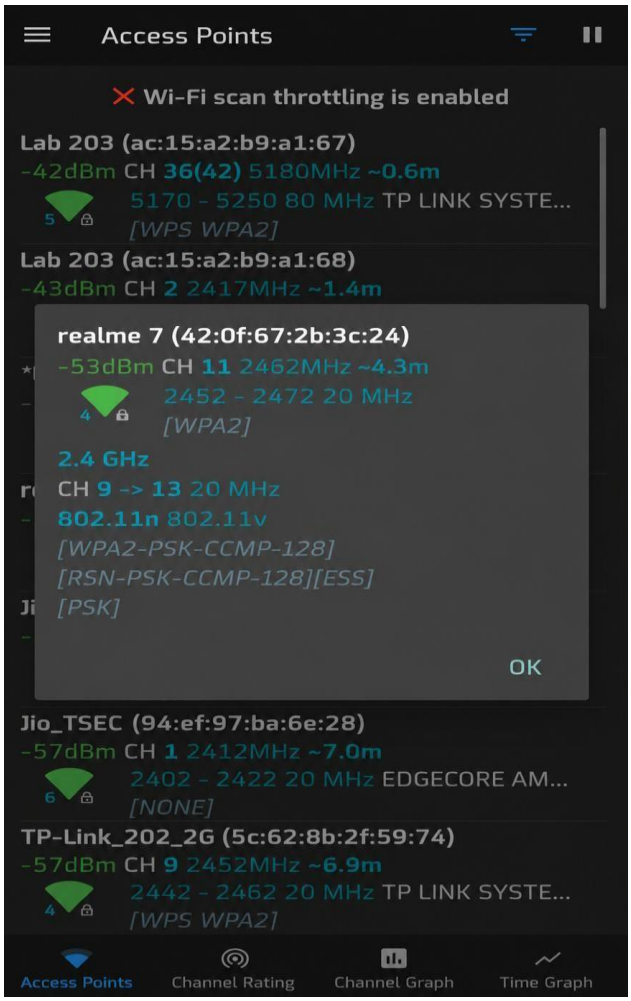
Additional Access Points can provide better signal coverage in areas where the original Access Point has weak signal strength or dead zones.

10. References

- **Dr. A. B. Patil's WMC Theory Notes** – Module 4, Section 4.4: Pathloss and Shadowing.
- **Rappaport, T.S.** – *Wireless Communications: Principles and Practice*.
- **Schwartz, M.** – *Mobile Wireless Communications*, Cambridge University Press.

Screenshots:






Access Points
Wi-Fi scan throttling is enabled
realme 7 (42:0f:67:2b:3c:24)
-51dBm CH 11 2462MHz ~3.4m
2452 - 2472 20 MHz
[WPA2]
Lab 203 (ac:15:a2:b9:a1:67)
-54dBm CH 36(42) 5180MHz ~0.6m
realme 7 (42:0f:67:2b:3c:24)
-51dBm CH 11 2462MHz ~3.4m
2452 - 2472 20 MHz
[WPA2]
2.4 GHz
CH 9 -> 13 20 MHz
802.11n 802.11v
[WPA2-PSK-CCMP-128]
[RSN-PSK-CCMP-128][ESS]
[PSK]
OK
Jio_Faculty (96:ef:97:ba:6e:28)
-68dBm CH 1 2412MHz ~15.2m
2402 - 2422 20 MHz
[NONE]
OnePlus Nord CE 2 Lite 5G (9e:f3:4f:c9:cb:46)
-59dBm CH 9 2452MHz ~8.7m
2442 - 2462 20 MHz
[WPA2-PSK-CCMP-128][WPA2]

Name: Asmit Pandey

Roll no: 2401072

Batch: T21

Experiment no. 07

Aim: To study Bluetooth communication between two mobile devices by performing device pairing and testing the communication range.

Objectives:

- To understand the Bluetooth communication process.
- To pair two Bluetooth-enabled mobile devices.
- To transfer data using Bluetooth.
- To measure the effective communication range.
- To observe the effect of obstacles on Bluetooth signal strength.

Theory module & outcome Mapping

Theory Module: Wireless Networking Technologies
Theory Section: Bluetooth – short range communication
CO: CO3
LO: LO4

Theory:

Bluetooth is a short-range wireless communication technology that operates in the 2.4 GHz ISM (Industrial, Scientific, and Medical) band. It enables wireless data exchange between nearby devices such as smartphones, laptops, headphones, smart watches, and IoT devices.

Features

- **Frequency Band:** 2.4 GHz
- **Standard:** IEEE 802.15.1
- Low power consumption
- Secure communication through pairing
- Supports voice and data transmission

Bluetooth Classes

Class	Maximum Power	Typical Range
Class 1	100 mW	Up to 100 m
Class 2	2.5 mW	About 10 m
Class 3	1 mW	About 1 m

Most smartphones use Class 2 Bluetooth, providing a practical range of approximately 10 meters.

Equipment Required

- Two Bluetooth-enabled smartphones
- Measuring tape
- Open indoor/outdoor area
- Stopwatch (optional)
- Notebook for observations

Experimental Setup

Smartphone A ~~~~~~~~~~ Bluetooth Link ~~~~~~~~~~ Smartphone B

Distance = 1 m to 20 m

Procedure:

Step 1: Bluetooth Pairing

1. Turn ON Bluetooth on both smartphones.
2. Make both devices discoverable.
3. Search for nearby Bluetooth devices.
4. Select the other smartphone from the list.
5. Confirm the pairing request on both devices.
6. Verify that the devices are successfully paired.
7. Transfer a small file (photo or PDF) to ensure successful communication.

Step 2: Range Testing

1. Place both devices 1 meter apart.
2. Transfer a small file and note whether the transfer is successful.
3. Increase the distance gradually (2 m, 4 m, 6 m, 8 m, 10 m, etc.).

4. At each distance:
    - Send the same file.
    - Record whether the transfer succeeds.
    - Measure the approximate transfer time.
5. Continue increasing the distance until the connection fails.
6. Repeat the experiment with obstacles (wall, door, furniture) between the devices.

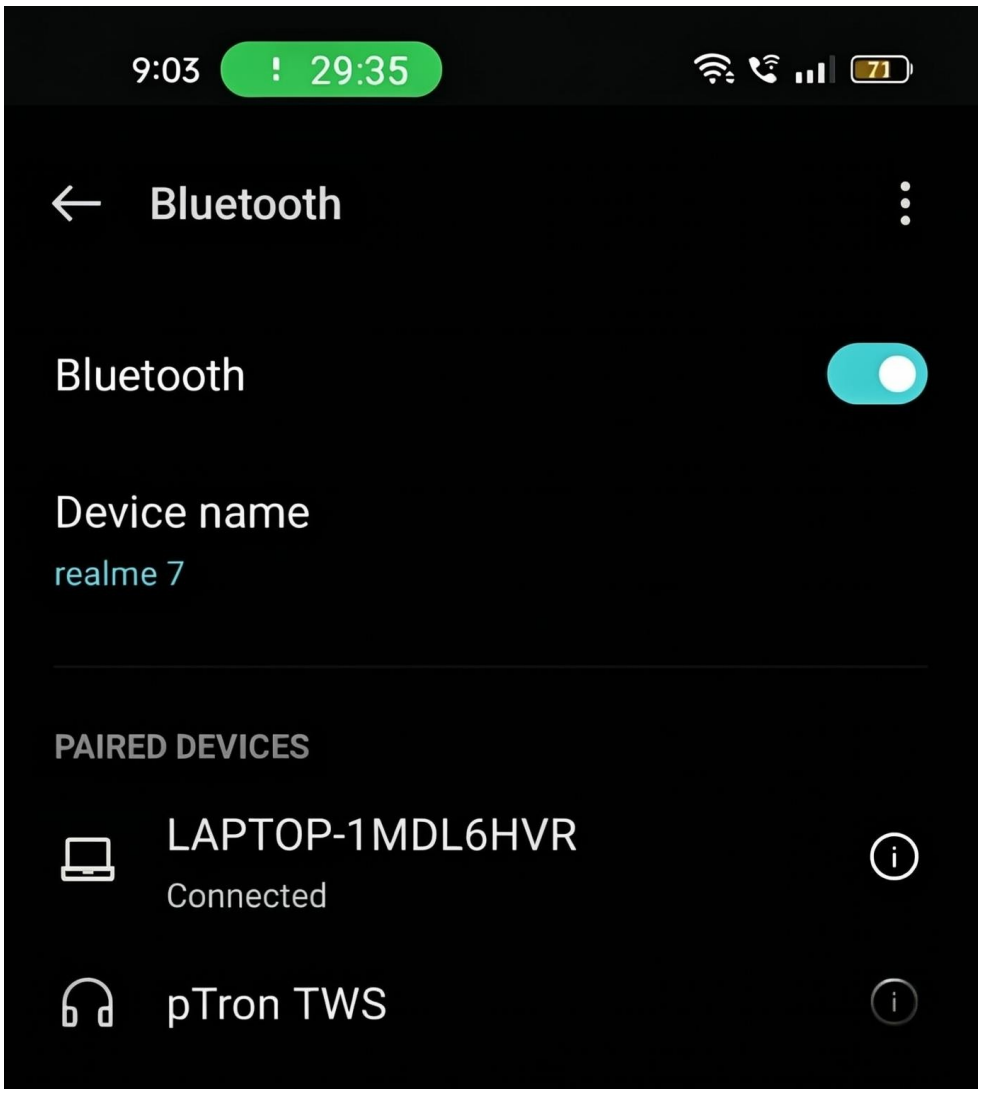
Observation Table

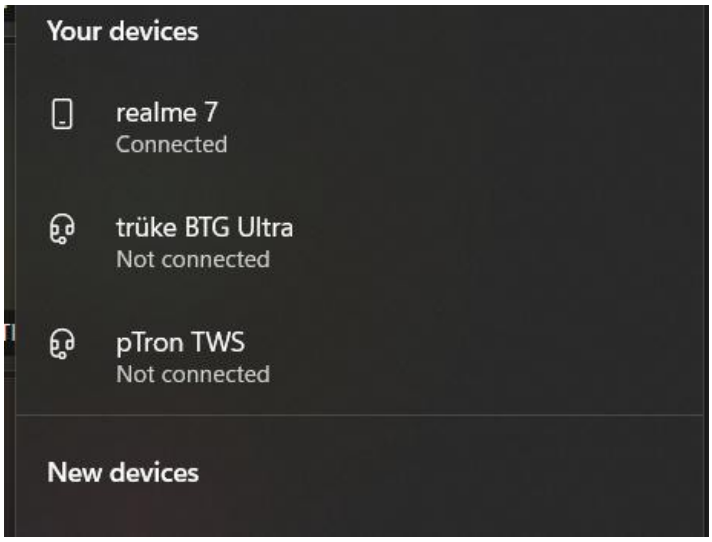
Open Space

Distance (m)	Pairing Status	File Transfer	Transfer Time (s)	Remarks
1	Successful	Successful	3	Good
2	Successful	Successful	3	Good
4	Successful	Successful	4	Good
6	Successful	Successful	5	Good
8	Successful	Successful	6	Slight delay
10	Successful	Successful	8	Slight delay
12	Successful	Failed	—	Weak signal
15	Successful	Failed	—	Out of range

With Wall Obstacle

Distance (m)	File Transfer	Remarks
2	Successful	Good
4	Successful	Slight delay
6	Successful	Weak signal
8	Failed	Connection lost





Result:

Bluetooth communication was successfully established between two mobile devices through device pairing and file transfer. In open space, successful file transfer was observed up to approximately 10 m, while the connection became unreliable beyond this distance. With a wall

obstacle, successful file transfer was observed up to approximately 6 m, while the connection was lost at 8 m.

Conclusion:

The experiment demonstrated short-range wireless communication using Bluetooth. The two mobile devices were successfully paired and data was transferred using Bluetooth. It was observed that communication performance decreases as the distance between the devices increases. The presence of a wall obstacle further reduced the effective communication range.

Viva Questions

1. What is Bluetooth?

Bluetooth is a short-range wireless communication technology used to exchange data between nearby devices.

2. What frequency band does Bluetooth use?

Bluetooth operates in the **2.4 GHz ISM band**.

3. What is Bluetooth pairing?

Bluetooth pairing is the process of establishing a trusted relationship between two Bluetooth-enabled devices so that they can communicate with each other.

4. Why is pairing required?

Pairing is required to authenticate the devices and establish a trusted connection for communication.

5. What is the typical Bluetooth communication range?

The typical range of Class 2 Bluetooth is approximately **10 meters**.

6. What factors affect Bluetooth range?

Bluetooth range is affected by:

- Distance between devices
- Transmit power
- Obstacles such as walls
- Interference
- Device sensitivity
- Surrounding environment

7. What is Bluetooth discovery?

Bluetooth discovery is the process by which a Bluetooth device searches for and detects nearby discoverable Bluetooth devices.

8. What is the difference between pairing and connecting?

Pairing establishes a trusted relationship between two Bluetooth devices, whereas **connecting** establishes an active communication link between already paired devices.

9. What is the difference between Bluetooth Classic and Bluetooth Low Energy (BLE)?

Bluetooth Classic is commonly used for continuous data and audio communication, whereas Bluetooth Low Energy (BLE) is designed for low-power applications such as sensors and IoT devices.

10. How can the Bluetooth range be tested?

Bluetooth range can be tested by gradually increasing the distance between two paired devices and checking whether data can still be transferred successfully.

11. Name some applications of Bluetooth.

Applications of Bluetooth include:

- Wireless headphones
- Smart watches
- Smartphones
- Wireless keyboards and mice
- File transfer
- Hands-free communication
- IoT devices

12. What is the main objective of this experiment?

The main objective is to study Bluetooth communication between two mobile devices by pairing them, transferring data, measuring the communication range, and observing the effect of obstacles on communication.

Name: Asmit Pandey

Roll no: 2401072

Class: T2 Batch: T21

WMC Experiment no. 08

Aim: To study Bluetooth communication between two mobile devices by performing device pairing and testing the communication range.

Objectives:

- To understand the Bluetooth communication process.
- To pair two Bluetooth-enabled mobile devices.
- To transfer data using Bluetooth.
- To measure the effective communication range.
- To observe the effect of obstacles on Bluetooth signal strength.

Theory module & outcome Mapping

Theory Module: Wireless Networking Technologies
Theory Section: Bluetooth – short range communication
CO: CO3
LO: LO4

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Bluetooth is a short-range wireless communication technology that operates in the 2.4 GHz ISM (Industrial, Scientific, and Medical) band. It enables wireless data exchange between nearby devices such as smartphones, laptops, headphones, smart watches, and IoT devices.

Features

- Frequency Band: 2.4 GHz
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Most smartphones use Class 2 Bluetooth, providing a practical range of approximately 10 meters.

Equipment Required

- Two Bluetooth-enabled smartphones
- Measuring tape
- Open indoor/outdoor area
- Stopwatch (optional)
- Notebook for observations

Experimental Setup

Smartphone A ~~~~~~~~~~ Bluetooth Link ~~~~~~~~~~ Smartphone B

Distance = 1 m to 20 m

Procedure:

Step 1: Bluetooth Pairing

1. Turn ON Bluetooth on both smartphones.
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3. Search for nearby Bluetooth devices.
4. Select the other smartphone from the list.
5. Confirm the pairing request on both devices.
6. Verify that the devices are successfully paired.
7. Transfer a small file (photo or PDF) to ensure successful communication.

Step 2: Range Testing

1. Place both devices 1 meter apart.

2. Transfer a small file and note whether the transfer is successful.

3. Increase the distance gradually (2 m, 4 m, 6 m, 8 m, 10 m, etc.).

4. At each distance:

    - Send the same file. o Record whether the transfer succeeds. o Measure the approximate transfer time.

5. Continue increasing the distance until the connection fails.

6. Repeat the experiment with obstacles (wall, door, furniture) between the devices.

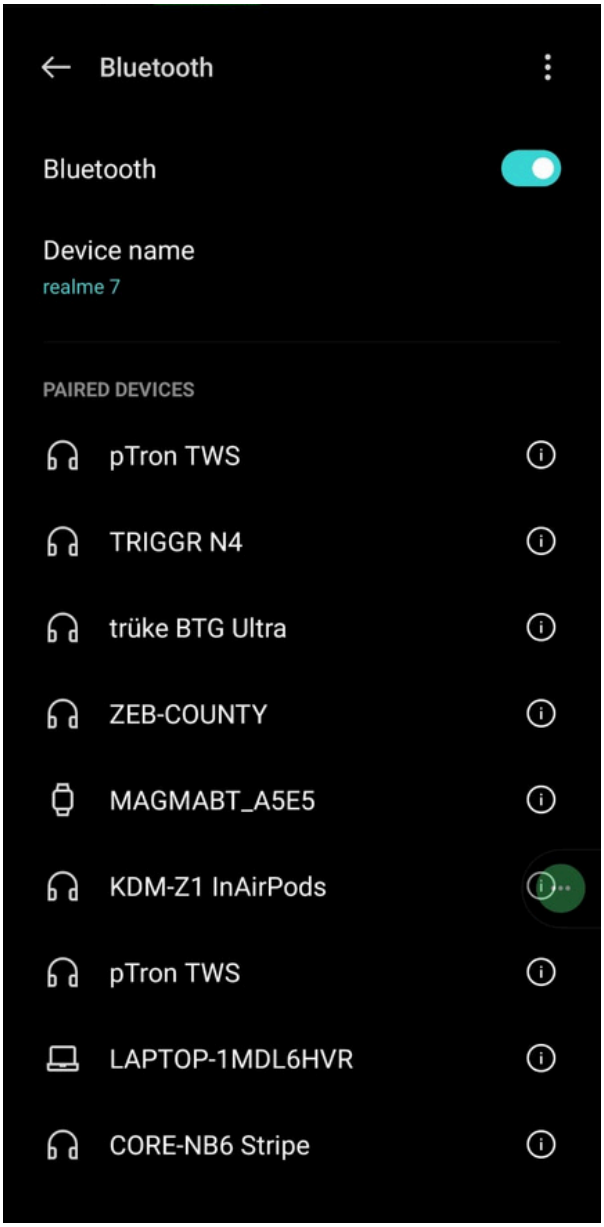
Observation Table

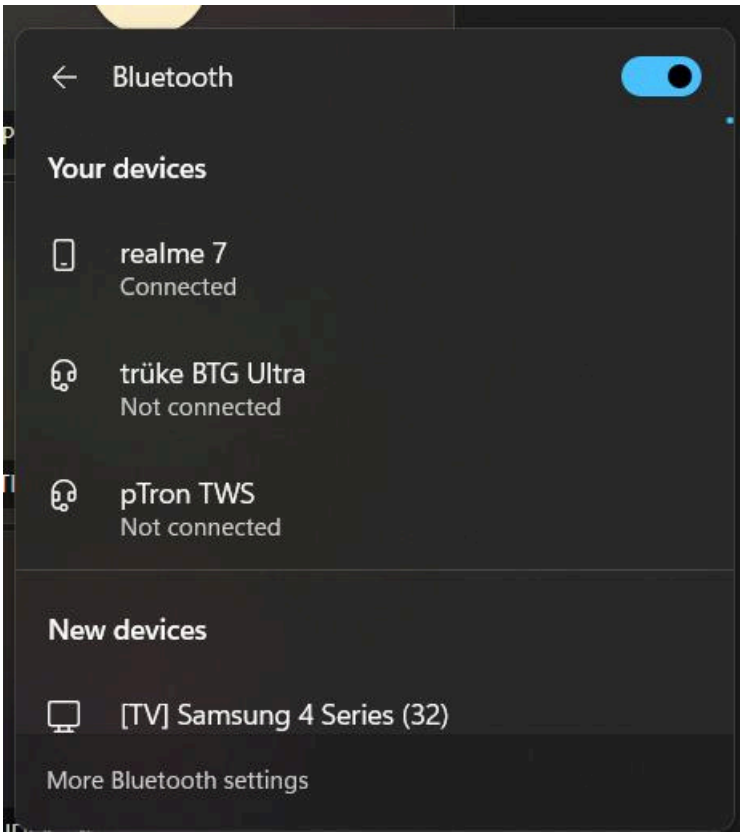
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15 Successful	Successful	8	Slight delay
Successful	Failed	—	Weak signal
Successful	Failed	—	Out of range

With Wall Obstacle

Distance (m) File Transfer	Remarks
2 Successful	Good
4 Successful	Slight delay
6 Successful	Weak signal
8 Failed	Connection lost


Bluetooth
Bluetooth
Device name
realme 7
PAIRED DEVICES
pTron TWS
TRIGGR N4
trüke BTG Ultra
ZEB-COUNTY
MAGMABT_A5E5
KDM-Z1 InAirPods
pTron TWS
LAPTOP-1MDL6HVR
CORE-NB6 Stripe



Result:

Bluetooth communication was successfully established between two mobile devices through device pairing and file transfer. In open space, successful file transfer was observed up to approximately 10 m, while the connection became unreliable beyond this distance. With a wall obstacle, successful file transfer was observed up to approximately 6 m, while the connection was lost at 8 m.

Conclusion:

The experiment demonstrated short-range wireless communication using Bluetooth. The two mobile devices were successfully paired and data was transferred using Bluetooth. It was observed that communication performance decreases as the distance between the devices increases. The presence of a wall obstacle further reduced the effective communication range.

Viva Questions

1. What is Bluetooth?

Bluetooth is a short-range wireless communication technology used to exchange data between nearby devices.

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Bluetoothenabled devices so that they can communicate with each other.

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11. Name some applications of Bluetooth.

Applications of Bluetooth include:

- Wireless headphones
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- Smartphones
- Wireless keyboards and mice
- File transfer
- Hands-free communication
- IoT devices

12. What is the main objective of this experiment?

The main objective is to study Bluetooth communication between two mobile devices by pairing them, transferring data, measuring the communication range, and observing the effect of obstacles on communication.

Name:Asmit Pandey

Roll no: 2401072

Class: T2 Batch: T21

Experiment No. 9

Title:

Study of Cellular Network Parameters Using a Smartphone (field observation).

1. Aim

To observe and analyze cellular network parameters such as**network type and signal strength** at different locations using a smartphone.

Specific Objectives:

- To identify the cellular network type used by a smartphone.
- To observe the received cellular signal strength.
- To compare cellular network parameters at different locations.
- To relate variations in signal strength and network type to cellular coverage and wireless propagation.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 6: Future Trends in Wireless Communication
Theory Section(s)	Cellular Communication / 4G and 5G
Course Outcome	CO6
Laboratory Outcome	LO6

3. Theoretical Background

3.1 Introduction

A cellular network provides wireless communication over a large geographical area by dividing the area into smaller regions called cells. Each cell is served by a cellular base station, which provides radio communication to mobile devices within its coverage area.

A smartphone communicates with the cellular network through a suitable cellular base station. As the user moves from one location to another, the received signal strength may change due to distance from the serving cell, buildings, obstacles and other wireless propagation conditions. The smartphone may also use different cellular technologies such as 2G, 3G, 4G/LTE or 5G, depending on network availability and operating conditions.

Therefore, observing cellular parameters at different locations provides a practical understanding of cellular coverage, signal strength and mobile connectivity.

3.2 Cellular Signal Strength

Signal strength indicates the strength of the cellular radio signal received by the smartphone. Depending on the smartphone and network technology, it may be displayed in dBm or as signal bars. When signal strength is displayed in dBm, the value is generally negative. A value closer to zero represents a stronger received signal. For example, a signal strength of −65 dBm is stronger than −90 dBm.

3.3 Cellular Network Type

A smartphone may connect to different generations of cellular technology depending on network availability and operating conditions.

Network Type

	Generation
2G	Second Generation
3G	Third Generation
4G / LTE	Fourth Generation
5G	Fifth Generation

The experiment does not require students to obtain all four network types. Students should record the network type actually being used by their smartphone at each observation location.

4. Requirements

1. Smartphone with an active SIM card.
2. Active cellular network connection.
3. Mobile Network / Mobile Data enabled.
4. Smartphone settings providing cellular network information.

5. Procedure

Step 1: Prepare the Smartphone

Ensure that the smartphone has an active SIM card and that Mobile Network / Mobile Data is enabled.

Step 2: Open Mobile Network Information

Open the smartphone settings and select the active SIM. Depending on the smartphone manufacturer and Android version, the menu may be available under:

Settings → About phone → Status information → SIM Card status

Step 3: Observe the Network Type

Open SIM Status / Network Status / Mobile Network Information and identify the current network type being used by the smartphone. Record whether the phone is currently using 2G, 3G, 4G/LTE or 5G, as displayed by the smartphone.

Step 4: Observe Signal Strength

Note the cellular signal strength. If the smartphone displays the signal strength in dBm, record the value. For example: Signal Strength = −72 dBm.

***Note*:** The exact location and terminology of *Network Type* and *Signal Strength* information may differ with the smartphone manufacturer and Android version. Students should use the corresponding SIM Status / Mobile Network / About Phone information available on their smartphone.

Step 5: Record the Observation at Home

At a suitable location inside the home, repeat Steps 3 and 4 and record the network type and signal strength.

Step 6: Record the Observation at a Railway Station / Public Place

At a railway station or another suitable public place, repeat Steps 3 and 4 and record the network type and signal strength.

Step 7: Record Observations During Travel

During travel, select two different locations and repeat Steps 3 and 4. Record the network type and signal strength at both locations. The two travel locations may be selected according to the student's normal travel route.

Step 8: Record the Observation at College Laboratory

At the college laboratory, repeat Steps 3 and 4 and record the network type and signal strength.

Step 9: Compare the Observations

Compare the network type and signal strength obtained at all five locations. Identify the locations where the signal was relatively stronger or weaker and note any change in network type.

***Important*:** Do not change the preferred network type manually. Record the network type actually being used by the smartphone at each location.

6. Observations

Sr. No.	Observation Location	Network Operator	Network Type	Signal Strength (dBm)
1	Home	Jio True5G-Jio	NR	-98 dBm 36asu
2	Railway Station / Public Place	Jio True5G-Jio	NR	-87 dBm, 34asu
3	Travel Location 1	Jio True5G-Jio	NR	-94 dBm, 35asu
4	Travel Location 2	Jio True5G-Jio	NR	-101 dBm 37asu
5	College Laboratory	Jio True5G-Jio	NR	-103 dBm 37asu

7. Analysis and Discussion

1. Compare the signal strength observed at the five locations. At which location was the cellular signal strongest and at which location was it weakest?

Since a dBm value closer to zero represents a stronger received signal, the Railway Station / Public Place showed the strongest observed signal at -87 dBm.

2. Did the network type remain the same at all five locations? If not, state the observed changes.

Yes. The network type remained the same at all five observation locations.

The smartphone was connected to NR (5G) through the Jio True5G-Jio network at all the observed locations.

No change in network type was observed during the experiment.

3. Why can cellular signal strength vary from one location to another?

Cellular signal strength can vary due to several factors, including:

- Distance from the serving cellular base station.
- Buildings and walls.
- Trees and other physical obstacles.
- Indoor and outdoor conditions.
- Radio interference.
- Reflection, diffraction and scattering of radio waves.
- Changes in the propagation environment.
- Network loading and radio conditions.

Therefore, the same smartphone can show different signal strengths at different locations.

4. How can distance, buildings and other obstacles affect cellular signal strength?

As the distance between the smartphone and the cellular base station increases, the received signal generally becomes weaker because of propagation loss.

Buildings, walls, concrete structures, trees and other obstacles can absorb, reflect, diffract or scatter radio waves. These effects can reduce the strength and quality of the signal received by the smartphone.

For example, the relatively weaker reading in the College Laboratory (-103 dBm) may be associated with indoor walls and other obstructions along with the prevailing radio propagation conditions.

5. What could be the possible reasons for a change in network type observed at different locations?

A smartphone may change its network type when the availability or quality of a particular cellular technology changes.

Possible reasons include:

- Weak or unavailable 5G coverage.
- Stronger availability of 4G/LTE in another area.
- Distance from the serving cell.
- Buildings and other obstacles.
- Network configuration.
- Radio propagation conditions.
- Movement of the user between different coverage areas.

In this experiment, however, no change in network type was observed, and the phone remained connected to NR (5G).

9. Result and Conclusion

Result

The cellular network parameters were observed at five different locations using a smartphone. The network operator was Jio True5G-Jio, and the observed network type was NR (5G) at all five locations.

The recorded signal strength ranged from -87 dBm to -103 dBm.

The strongest observed signal was -87 dBm at the Railway Station / Public Place, while the weakest observed signal was -103 dBm at the College Laboratory.

Conclusion

The experiment demonstrated that cellular signal strength varies from one location to another due to factors such as distance from the cellular base station, buildings, obstacles and wireless propagation conditions.

The smartphone remained connected to NR (5G) at all five observation locations, although the received signal strength changed. Thus, the experiment provided practical understanding of cellular coverage, signal strength, network type and wireless propagation.

10. Viva Questions

Q1. What is a cellular network?

A cellular network is a wireless communication system in which a geographical area is divided into smaller regions called cells, each served by a cellular base station.

Q2. What is a cell in cellular communication?

A cell is a geographical area covered by a particular cellular base station.

Q3. What is meant by cellular signal strength?

Cellular signal strength is the strength of the radio signal received by a mobile device from the cellular network. It is commonly expressed in dBm.

Q4. Why can signal strength change when a user moves from one location to another?
Signal strength can change because the distance from the base station and the surrounding propagation environment change as the user moves.

Q5. What is the role of a cellular base station?

A cellular base station provides the radio communication link between mobile devices and the cellular network.

Q6. Why may a smartphone change its network type while travelling?

The smartphone may change network type because the availability and quality of different cellular technologies vary between different coverage areas.

Q7. How do buildings and other obstacles affect cellular signals?

Buildings and obstacles can absorb, reflect, diffract and scatter radio waves, which can reduce the received signal strength.

11. References

- **Dr. A. B. Patil's WMCTheoryNotes** – Cellular Communication, 4G and 5G.
- **Theodore S. Rappaport** – Wireless Communications: Principles and Practice, Pearson**.**
- **William Stallings** – Wireless Communications and Networks, Pearson.

THADOMAL SHAHANI ENGINEERING COLLEGE
DEPARTMENT OF

ArtificialIntelligence&DataScience

Name of Student	:	**Asmit Pandey**
Class	:	**TE**
Division	:	**T2**
Roll Number	:	**2401072**
Batch	:	**T21**
Subject	:	**Wireless and Mobile Communication Lab**
Experiment Number	:	**10**
Experiment Title	:	**Introduction to Network Simulation Using Cisco Packet Tracer – Smart Home Wireless IoT Network.**
Date of Performance	:	**13-10-2026**
Date of Submission	:	**13-10-2026**

Experiment No. 10

Title:

Introduction to Network Simulation Using Cisco Packet Tracer – Smart Home Wireless IoT Network.

1. Aim

To use a GUI-based network simulation tool, Cisco Packet Tracer, to design, configure and test a basic wireless IoT network for a smart home application.

Specific Objectives:

- To create a basic wireless IoT network using Cisco Packet Tracer.
- To connect IoT sensors and actuators through a wireless Home Gateway.
- To configure a simple automation rule between a sensor and actuators.
- To observe the operation of the simulated wireless IoT network.

2. Theory Module & Outcome Mapping

Item	Mapping
Theory Module	Module 3 – Wireless Networking Technologies
Theory Section(s)	Wi-Fi Technology
Course Outcome	CO3
Laboratory Outcome	LO6

3. Theoretical Background

3.1 Network Simulation

Network simulation is the process of creating a virtual model of a computer or communication network using software. It allows the behaviour of network devices and data communication to be studied without physically building the complete network. A network simulator provides virtual devices such as routers, switches, wireless access points, computers and IoT devices. User can configure these devices, establish connections and observe network operation.

3.2 Cisco Packet Tracer

Cisco Packet Tracer is a GUI-based network simulation tool that allows users to create and test network topologies using virtual devices. Devices can be placed and connected using a drag-anddrop interface. It provides Realtime Mode as well as Simulation Mode to observe packet movement through the network.

3.3 Wireless IoT Network

An Internet of Things (IoT) device is a physical device equipped with sensing, processing and communication capabilities that can exchange data over a network. In a smart home, sensors can detect conditions such as motion, while actuators can perform actions such as switching ON a light, fan, appliance, siren etc.

In this experiment, a Home Gateway provides wireless connectivity to IoT devices. A Motion Detector acts as a sensor, while a Siren act as actuators.

The basic automation is represented as:

IF Motion is detected → THEN Siren is activated.

4. Requirements

1. Laptop or desktop computer.
2. Cisco Packet Tracer software with IoT support.

5. Procedure

Step A: Setting up the Home Gateway (Wireless Access Point)

1. Open Cisco Packet Tracer and create a new workspace.
2. Drag a Home Gateway from Network Devices > Wireless Devices onto the workspace.
3. Click on the Home Gateway, open the Config tab, and select Wireless.
4. Set the SSID to SmartHome_IoT, select WPA2-PSK security, and enter a secure passphrase (e.g., TSEC@123).
5. Note the default Gateway IP address (e.g. 192.168.25.1), which serves as the local web server address. (Select LAN in Config tab to find Gateway IP address mentioned in IPv4 address.)

Step B: Deploying and Connecting IoT End-Devices

1. Place a Motion Detector and Siren from the End Devices > Home category onto the workspace.
2. Click on each device, navigate to Config > Wireless, enter the SSID to SmartHome_IoT, and enter the correct WPA2-PSK passphrase.
3. Verify that each device associates successfully with the Home Gateway and obtains an IP address via DHCP in the 192.168.25.x subnet.

Step C: Configuring IoT Server Registration

1. Click on each IoT end-device (Motion Detector and Siren), go to its Config > Settings tab, and locate the IoT Server setting.

2. Change the setting from None or Home Gateway to Remote Server.

3. Enter the Home Gateway server IP address (which you have noted in step A, e.g. 192.168.25.1) along with default server credentials (admin / admin). This registers the device state parameters into the central gateway database.

Step D: Accessing the Central Management GUI via Smartphone

1. Place a Smartphone on the workspace and configure its wireless interface Config > Wireless, enter the SSID as SmartHome_IoT, and enter the correct WPA2-PSK passphrase.
2. Open the Smartphone, go to the Desktop tab, and launch the Web Browser application.
3. Enter the Home Gateway IP address (which you have noted in Step A e.g. 192.168.25.1) in the URL bar and log in with the administrator credentials (admin / admin).
4. The IoT Monitor dashboard loads, displaying active connected devices (Motion Detector and Siren).

Step E: Establishing Conditional Automation Rules (IF-THEN Logic)

1. Log In via the Smartphone Browser: Open the Smartphone on your workspace, launch the Web Browser application, and enter the Home Gateway IP address (e.g. 192.168.25.1) in the URL bar. Log in using the administrator credentials (admin / admin). (Verification: The IoT Monitor dashboard loads successfully on the smartphone screen, showing active connected nodes.)

2. Navigate to Conditions and Create a Rule: Click on the Conditions link at the top right of the IoT Monitor interface, then click the Add button to open the "Add Rule" configuration window. (Verification: The rule builder form appears, displaying the Name text box, Enabled checkbox, and conditional logic blocks.)

3. Configure the Motion Detector Input: Type Test01 in the Name field and ensure the Enabled checkbox is checked. Under the If clause, select your Motion Detector (e.g. IoT0) from the first dropdown menu, choose the property parameter (On) from the second dropdown, and set its matching state to True.

4. Configure the Siren Output: Under the Then set clause, select the Siren (IoT1), and set its target state On is True.

5. Save the Automation Logic: Click the OK button at the bottom right corner of the rule window to save the automation logic directly to the Home Gateway server database. (Verification: The window closes and the newly created rule populates into the main IoT Monitor table list Press Ok.)

Step F: Testing and Verification

1. Real-Time Trigger Test: Hold down the Alt key on your keyboard, place your cursor over the Motion Detector in the workspace to simulate physical movement.

2. State Activation Observation: Observe that the motion detector changes state to True, instantly executing the conditional rule to command the Siren to activate.

3. Deactivating Actuators: Once the sensor resets (generally after five seconds), manually turn off the Siren through the holding down the Alt key and by clicking their physical icon, returning the system to its initial baseline state.

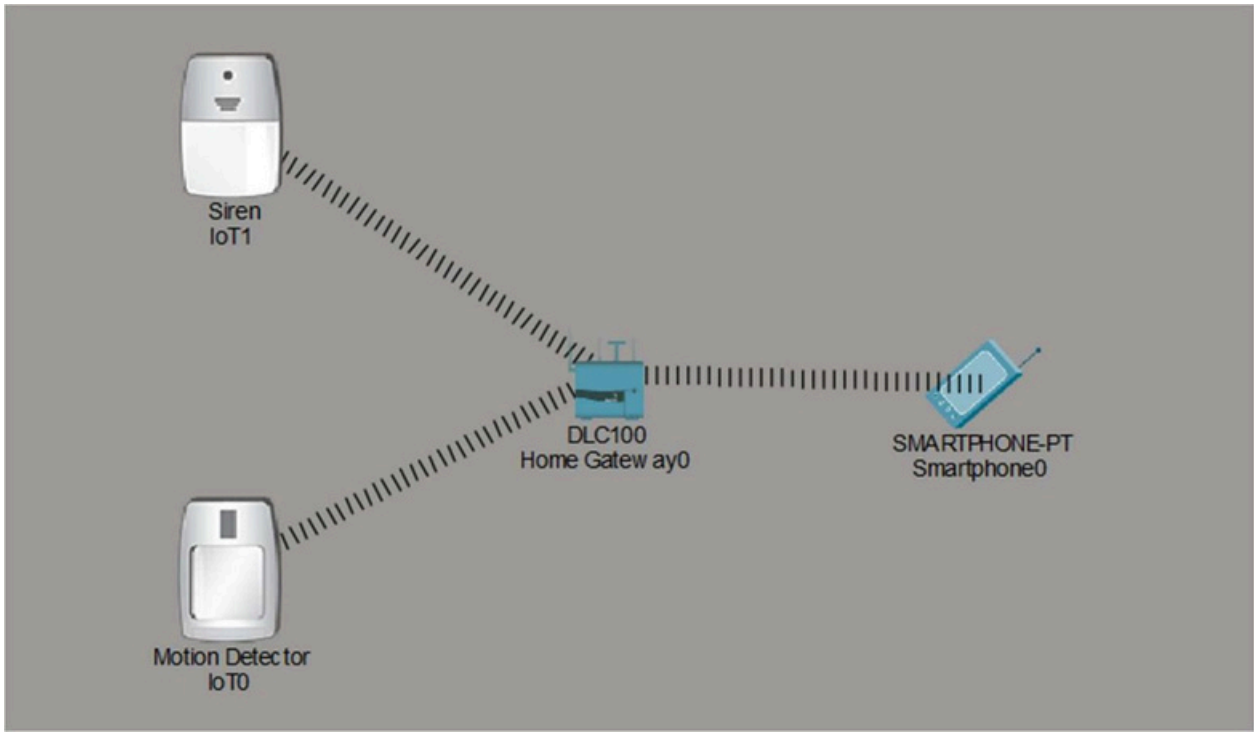
4. Iterative Verification: Repeat the Alt + Click action on the motion detector to verify that the automated loop can be triggered consecutively without errors.

6. Observations

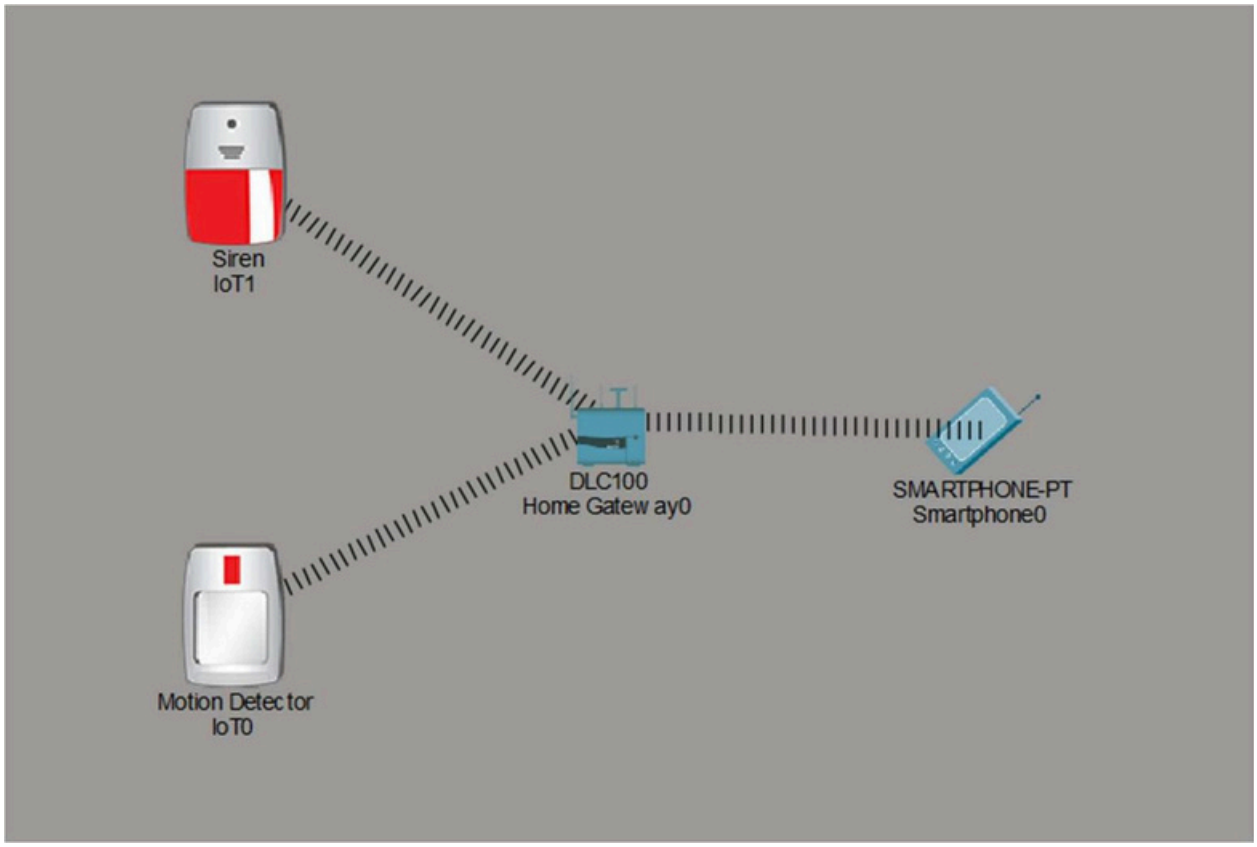
Device Name	Device Type	IP Address	Operational State After Trigger
Motion Detector	Input Sensor	192.168.25.101	Active (Motion Detected = True)
Siren	Output Actuator	192.168.25.102	ON (Alarm Sounding)

7. Screenshots

a) Initial Condition



b) Activated Condition



8. Task 1 - Smart Home with Two-Room Automation

Design and simulate a two-room smart home in Cisco Packet Tracer. The smart home consists of Room A and Room B. Each room shall contain: One Motion Detector, One Smart Light and One Smart Fan. Configure the system so that:

Room A: IF Motion Detector A = ON → THEN Smart Light A = ON AND Smart Fan A = ON
Room B: IF Motion Detector B = ON → THEN Smart Light B = ON AND Smart Fan B = ON
Device Naming Requirement: Change the default display names of all IoT devices from names such as IoT0, IoT1, etc. to meaningful names using the following format:

DeviceName_RollNumber_Room

where the room is represented by A or B.

For example, for a student with roll number 12, a Smart Fan in Room A shall be named as

Fan_12A. Use the last two digits of your roll number in the device name.

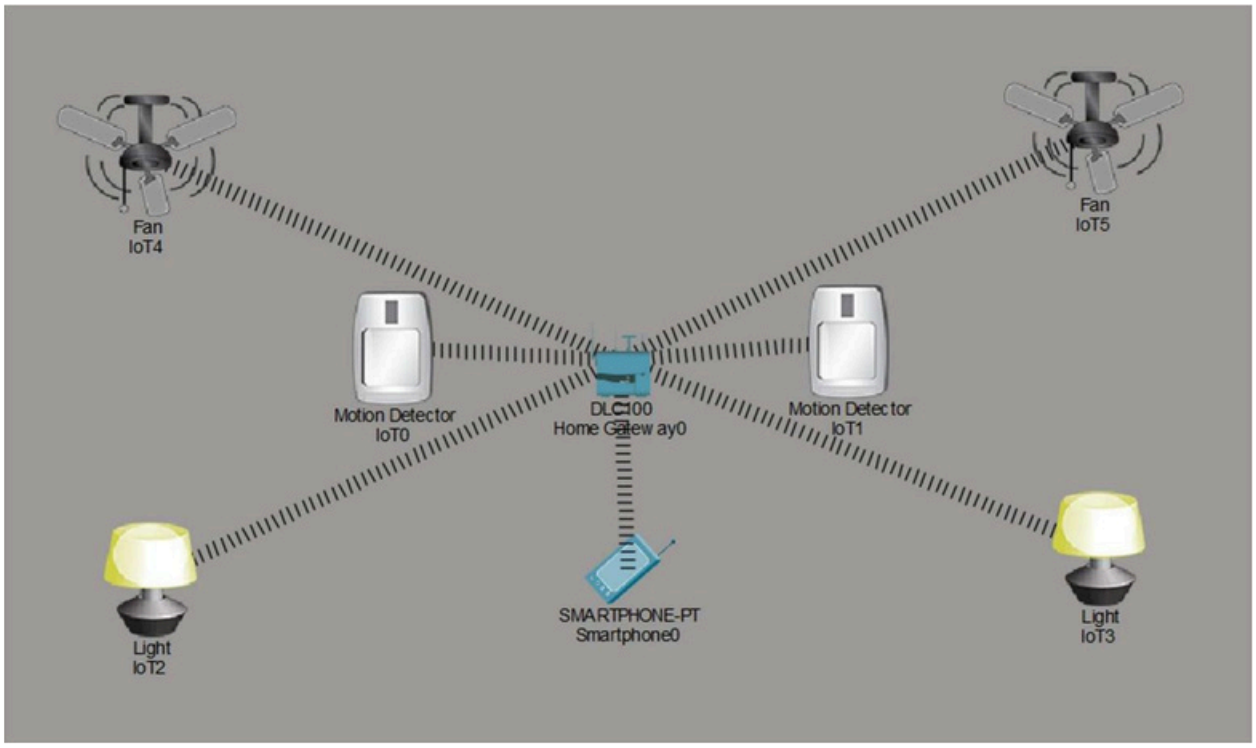
Submission Guideline

Record a 12–15 second video demonstrating the working of the simulated smart home.

The video should show:

1. The initial condition, with the Smart Fans and Smart Lights in both rooms switched OFF.
2. Motion being detected in Room A, followed by automatic activation of the Smart Fan and Smart Light in Room A.
3. Motion being detected in Room B, followed by automatic activation of the Smart Fan and Smart Light in Room B.

The device display names should be clearly visible in the video.



9. Result and Conclusion

In this experiment, a wireless smart home IoT network was successfully designed, configured, and tested using Cisco Packet Tracer. The Home Gateway provided seamless Wi-Fi connectivity and DHCP address allocation to all end-devices. By setting up conditional IF-THEN automation rules on the Home Gateway central server, the Motion Detector effectively triggered the Siren upon detecting activity. The experiment successfully demonstrated the principles of wireless sensor network integration and automated IoT control.

10. Viva Questions

Q1. What is network simulation?

Ans: Network simulation is a software-based technique used to model and analyze the behavior of computer networks, devices, and communication protocols in a risk-free virtual environment without requiring real physical hardware.

Q2. What is the role of the Home Gateway in this experiment?

Ans: The Home Gateway serves as a central wireless access point, DHCP server, and local web server. It provides Wi-Fi connectivity to IoT devices and hosts the central management GUI where conditional automation rules are stored and executed.

Q3. What is the difference between a sensor and an actuator?

Ans: A sensor measures and detects physical inputs from the environment (e.g., motion, temperature) and converts them into electrical signals. An actuator receives electrical signals or control commands and converts them into physical action or output (e.g., sounding an alarm, switching on a light or fan).

Q4. What is the role of the Motion Detector in this experiment?

Ans: The Motion Detector acts as the primary input sensor that monitors human or physical movement and sends state updates (True/False) to the Home Gateway to trigger configured automation rules.

Q5. What would happen if the Motion Detector of Room A is activated?

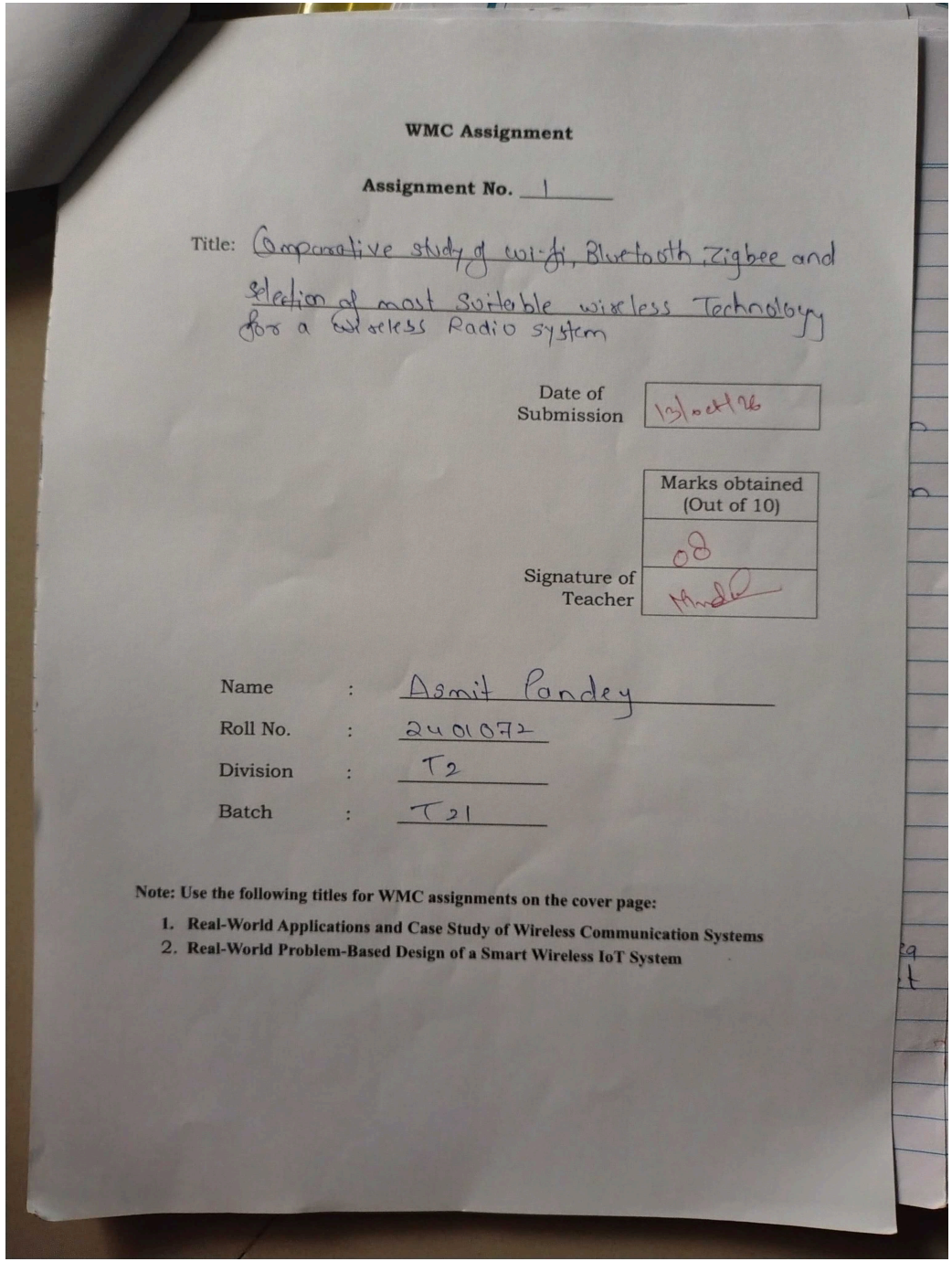
Ans: Activating the Motion Detector of Room A triggers its specific IF-THEN rule on the server, automatically switching ON both Smart Light A and Smart Fan A, while leaving Room B devices unaffected.

Q6. What is the purpose of using Realtime Mode in this experiment?

Ans: Realtime Mode allows network events and automation rules to execute in real time as they would in a physical network, enabling immediate feedback when triggering sensors or interacting with devices via the web interface.

11. References

- Dr. A. B. Patil's WMC Theory Notes – Module 3, Wireless Networking Technologies.
- Cisco Packet Tracer – IoT and Smart Home simulation resources.



WMC Assignment

Assignment No. 1

Title: Comparative study of wi-fi, Bluetooth, Zigbee and selection of most suitable wireless Technology for a wireless Radio system

Date of Submission	13/oct/26

Marks obtained (Out of 10)
08

Signature of Teacher: Mandal

Name : Amit Pandey

Roll No. : 240172

Division : T2

Batch : T21

Note: Use the following titles for WMC assignments on the cover page:

1. **Real-World Applications and Case Study of Wireless Communication Systems**
2. **Real-World Problem-Based Design of a Smart Wireless IoT System**

Name: Asmit Pandey
Batch: T21
Roll No: 2401072
Subject: WMC

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Assignment - 1

To study & compare wifi, Bluetooth and zigbee & select most suitable wireless technology for a given real world application

Introduction

Wireless communication is transfer of information between electronic devices without using physical cables. It is widely used in modern applications such as smart homes, offices, healthcare systems & IOT applications

3 Important wireless techlogy.

1] WiFi
2] Bluetooth
3] Zigbee

Part A - Comparative study

Task 1: Brief Note on Wireless technology

1] Wi-fi

- primary purpose: Designed for wireless local Area Network to provide high-speed wireless internet & local network access.

- Typical operating range: Approximately 50 to 100 m

Data Rate characteristics : High data rates ranging from 54 mbps in older standards to several Gbps

- Power Consumption : High Power Consumption, requires regular charging

- Typical Consumption : Home & enterprise network video streaming, large file transfers

ii) Bluetooth :

- Primary purpose : Designed for wireless Personal Area Network as a short range cable

- Typical operating Range : around meters

- Data Rate characteristics : medium data rates generally between 1 to 3 mbps.

- Power Consumption : low to medium also ultra low power consumption for background syncing

- Typical Applications
  wireless audio, connecting smartwatches to phones.

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iii) Zigbee

- Typical Purpose: Designed for Low-Rate wireless personal area Network.
- Typical operating range
  10 to 100 meters, though effective range is heavily extended via mesh networking
- Data rate: Low data rates. i.e. 250 kbps
- Power Consumption
  Very Low, allowing battery operated sensors to run for years
- Typical applications - Smart home automation (smart lighting, thermostats, door locks)

Task 2: Comparative Analysis

Parameter	Wifi	Bluetooth	Zigbee
Primary Purpose	WLAN	WPAN / cable replacement	LR-WPAN / control
operating range	2.4 GHz 5 GHz	2.4 GHz ISM	2.4 GHz

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Data Rate	High (upto several Gbps)	1-3 Mbps	Several Kbps
Power consumption	High	Medium	Very low
Number of devices	upto 250	7 active devices	upto 65000
Application	video streaming	wearables	IOT sensor
major advantages	High power, large capacity	easy pairing	low power reliable
major limitator	Complex Setup	limited range interference	low data rate, requries central hub

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Task 3 - Analysis

Which technology is more suitable for high speed, internet acess? why?

Wifi, it is only technology among the three capable of delivering high data rates required for modern internet usage like video streaming and heavy downloads

Which technology is most suitable for connecting nearby personal devices? Why?

Bluetooth, it is specially designed for wireless personal Area Networks. It provides a seamless pairing process, adequate data rates for audio & peripheral communication and is natively integrated into virtually all smartphones.

Which technology is most suitable for a low power sensor network? why?

Zigbee, it was purpose built for low data rate, lower power applications. Its ability to form large, self healing mesh networks allows thousands of sensors to communicate over wide areas.

Part B - Application Based T
Selection

Task 4:
Scenario 2- Wireless Personal Devices
A user wants to connect a smartphon
with wireless headphone a smart wa
& other nearby devices.

Scope Application:
Scenario involves establishing a localise
networks centred around single highly mob
user. The primary engineering challeng
is routing data between a central hub
c the smartphone and multiple wearable
or portable peripherals within an
extremely close physical proximity. The
analysis requires evaluating, wifi, bluetooth
& zigbee to identify.

Task 5: Communication Requirement Analysis

Requirement	Requirement for Application
i) Required communication Range	short (0-10 meters, as devices are worn on or kept near body)

Requirement		Medium (Sufficient for continuous audio streaming)
3] Power Requirement		low (wearables & headphones have small)
4] Number of devices		few
5] Type of information		Audio streams (music/calls), health telemetry.
6] Important Application Requirement		Seamless pairing, Secure connection & low latency for audio video sync.
7] Mobility Requirement		High

Task 6: Technology Selection

The most suitable wireless technology for scenario 2 is bluetooth.

Justification:
Communication Range & mobility
Bluetooth is explicitly designed for short range WPAN applications

- Data Rate and Type of Information
Bluetooth classic provides the necessary medium data rate to steam high quality

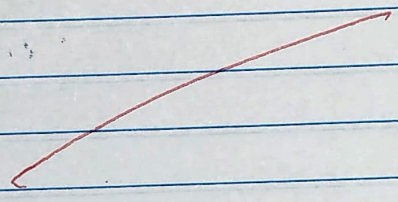
- Power consumption
Bluetooth Low Energy operates on ultra low power which is critical for a Smartwatch that needs to maintain a constant background connection for health telemetry and notifications without rapidly draining its small battery

- Number of Devices & Internet Connectivity
A smartphone can easily act as the master device in a bluetooth Piconet, managing a few slave devices. Because Bluetooth relies on Smartphone for heavy processing and internet routing the peripherals

Task 7: Why other technology is less suitable

- Why wifi is less suitable:
wifi is designed for high bandwidth & long range networking, which comes at cost of high power consumption. If wireless headphones or a smartwatch used in wifi

- Why zigbee is less suitable
Zigbee has a very low maximum data rate which is entirely insufficient streaming continous, high-quality audio to wireless headphones. Additionally, zigbee is designed for fixed mesh networks rather than highly mobile personal networks. Most importantly consumer smartphones do not have built in zigbee radios, meaning the user would needy to carry an impractical external to connect their devices.



WMC Assignment

Assignment No. 2

Title: Mini project on Smart Healthcare monitoring System

Date of Submission	13/oct/26
Marks obtained (Out of 10)	08
Signature of Teacher	Mhatre

Name : Asmit Pandey

Roll No. : 240172

Division : T2

Batch : T21

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1. **Real-World Applications and Case Study of Wireless Communication Systems**
2. **Real-World Problem-Based Design of a Smart Wireless IoT System**

Asmit Pandey
240102, T21
Subject: ~~HCS~~ WMC.

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Assignment 2 (8) 19/10/25 13/oct/25

Part A – Application and Problem Analysis

Task 1: Select Assigned Application Scenario:

the selected scenario is Scenario 2 - Smart Healthcare monitoring System. A healthcare facility requires a wireless system to collect information from patient monitoring or wearable devices and transmit the information to central monitoring station. The system must support communication between multiple monitoring devices and healthcare personal

Task 2: Application and Problem Analysis

1. Main purpose of proposed system:
The primary purpose is to wirelessly collect, transmit, and aggregate physiological data from patient wearable sensors to a central hospital dashboard for continous, real-time observation.

2. Problem the system solves:
It eliminates need for cumbersome wired monitoring equipment. allowing patients to move freely while ensuring healthcare staff can instantly detect and respond to critical medical events

Major devices or users involved
The system involves patient wearable sensors beside gateways or smartphones, wireless acess points, central server stations and healthcare personnel using viewing termina or mobile devices.

4. Why wireless Communication is suitable
Wireless communication is essential because it facilitates patient communication, reduces physical clutter in hospital wards, enables continous tracking across different hospital areas

Part B - Communication Requirement Analysis

Task 3: Identify the Communication Requirements

Requirement	Requirment for Assigned Application
Required Communication Coverage	Local Area and Personal Area
Number of Devices	High (multiple sensors per patient, scaled across many patients)

3) Data Rate Requirements	low / medium (continous small packets of vital signs)
4) Power Requirement	low (wearable sensors operate on small batteries)
5) Mobility Requirement	Both (Patients are mobile central monitoring station is stationary)
6) Type of Information	Physiological data (heart rate oxygen levels)
7) Requirment of Internet Connectivity	optional but recommended (for remote doctor access)
8) Reliability Requirment	High (critical health data requires near 100% uptime)
9) Important Environmental Challenges	Electromagnetic interference from medical equipment signal blocking by walls.

Task 4: Requirment Analysis

The most critical communication requirements for this system are high reliability and low power consumption. High reliability is param because a droped signal or delayed transmision of a critical vital sign ~~like~~ could result in missing a life threatning medical emergency low power consumption is equally critical because patient monitoring devices are typically wearables they must operate continously on small batteries without requiring frequent recharging. which would interupt monitoring process.

Part C - Wireless Technology Selection

Task 5: Select Suitable Wireless technology

To meet diverse requirement, a two ties hybrid wireless architecture is most suitable utilizing both Bluetooth low energy and wifi.

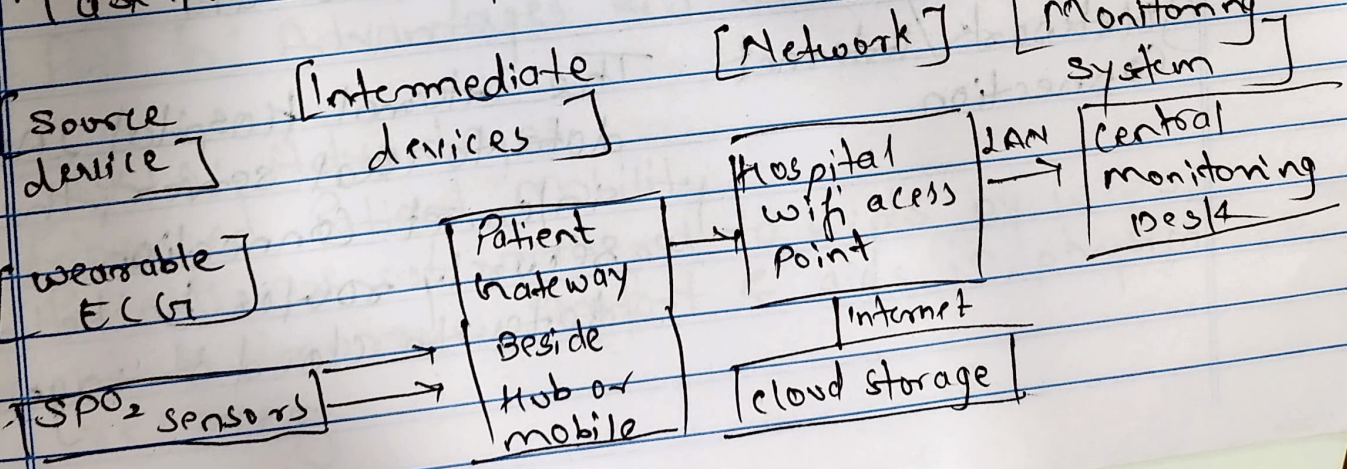
Task 6: Justification for technology selection

- Bluetooth Low Energy: BLE is selected for personal Area network to connect on-body sensors to a local patient gateway. It is ideal to its extremely low power consumption, which extends the battery life of wearables. Its data rate and communication range are perfect.

- wifi: wifi is selected for local area network to transmit aggregated data from the patient gateways to central monitoring station wifi provides necessary hospital wide communication range via access points, high data rates to handle multiplexed data from hundreds of patients

Part D - Proposed system Architecture

Task 7: Draw system Architecture



Part 5 - Description of System Components

Task 8: Explain function of each major Component

System Component	function in Proposed System
1] Source Device	wearable ECG, SpO_2 sensor and temp sensor continously measure patients physiological parameters.
2] wireless Communica Technology	BLE facilates low-power data transfer from sensors to local gateway, wifi transmits aggregated data across hospital network.
3] Gateway/Access points	Patient Gateway aggregates local BLE sensor data, the hospital wifi Access point receives this data
4] Network/Internet Connection	The Internal Hospital LAN routes data to local server, the internet Connection securely routes backed-up data to cloud storage.

Part F - Working of Proposed System

Task 1: Explain step-by-step Communica

1] Information Generation:
Wearable sensors continously monito- patient

2] Wireless Transmission
Sensors transmit these data packets via Bluet

3] Network Communication
Patient gateway aggregates data & transmits it
wifi to nearest hospital access point.

4] Processing
Central server receives data logs it into patien
electronic health record

5] Final user or Application
Processed data is pushed to user Application.

Task 2: Advantages and Limitations.

* Task 2a: Advantages
1. Enhanced Patient mobility
Patients are not tethered to beds by wires,
which improves comfort & aids in physical
recovery.

4] Rapid Emergency
Automated, real-time data transmission

5] Reduced Human Error
Continous automated logging prevents manual data entry errors.

Task 4: Limitations & possible solutions.

Limitation: hospital environment is prone to physical signal obstruction
Practical solution: Conduct a Comprehensive RF site survey prior to deployment & install a dense, overlapping mesh network.

Limitation: Wearable sensors have constrained battery capacities, continous transmission can deplete power quickly, causing monitoring interruptions.
Practical Solutions: Utilize aggresive duty-cycling in BLE transmission so radio is only active for ms.

Part 4 - Conclusion
Proposed Smart Healthcare Monitoring System